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Estimating maneuvering coefficients using system identification methods with experimental, system-based, and CFD free-running trial data

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ABSTRACT

Predicting ship maneuverings in calm water is an important topic; however, it is very expensive to run numerous maneuvering simulations using CFD. Despite its short computation time, system-based simulations need many captive model tests to estimate the hydrodynamic and rudder maneuvering coefficients used in the system-based mathematical model and the results have difficulty to achieve quantitative agreement with experimental data. Herein system identification techniques (extended Kalman filtering and constrained least square method using generalized reduced gradient algorithm) are used to predict the maneuvering coefficients from several EFD, systems based and CFD free-running trials. CFD gives not only the ship motions but also the total and component hydrodynamic forces/moments during the free-running simulations, which is helpful for estimating the maneuvering coefficients. Several types of free-running (turning circle, zigzag and large angle zigzag) tests are conducted using EFD and CFD to examine which free-running trial gives the best maneuvering coefficients using system identification. Also several free-running data are combined together for parallel processing. The set of maneuvering coefficients estimated by the constrained least square method using combined CFD free-running trial data show the most generalized results which covers a wide range of maneuvers.

1. Introduction

Ship transportation is increasing globally as is risk of collision especially in congested areas. Moreover, capsizing due to broaching is one of the most dangerous phenomena for ships maneuvering in waves. Ship maneuvering in calm water and waves are important topics to avoid collisions and broaching. Therefore reliable ship maneuvering simulations are required for incident analysis and prevention.

System-based (SB) and computational fluid dynamics (CFD) methods are major simulation methods to predict ship maneuverability. Computation time of the SB simulation is much shorter than that of CFD since such methods need only solve the equations of motion using a prescribed mathematical model and maneuvering coefficients. SB simulation requires approximately one minute for one free-running trial while CFD needs a few weeks or a month depending on the turbulence and propulsor modeling and size of grid. However, there are many issues in derivation of most accurate mathematical models and maneuvering coefficients, especially for wave conditions. In most cases captive model planar motion mechanism or rotating arm/circular motion tests are used, which require many static and dynamic conditions. On the other hand, CFD just requires the ship geometry and the propeller characteristics for the simulation (along with expert user and high performance computing resources).

Several SB models have been developed. Abkowitz (1964) developed a mathematical model to describe the hydrodynamic forces/moments acting on the ship with polynomial expressions using Taylor expansion on state variables. Christensen and Blanke (1986) developed 2nd order modulus expansions, which represent the hydrodynamic forces at angle of incidence: cross-flow drag. Recently a new maneuvering model was developed from first principles of low aspect-ratio aerodynamic theory and Lagrangian mechanics (Ross et al., 2007). Meanwhile the Maneuvering Mathematical Modeling Group (MMG) (Ogawa and Kasai, 1978; MMG, 1980) developed a mathematical model, which explicitly includes the individual open water characteristics of the hull/propeller/rudder and their interactions. The MMG model has shown good results for maneuvering in calm water (MMG, 1980; Stern et al., 2011) and qualitative results for extensions for wave conditions (Araki et al., 2010; Yasukawa, 2006). Issues for

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improvement include both wave terms in the mathematical model and methods for obtaining the wave maneuvering coefficients. Usually the maneuvering coefficients are assumed constant, which could be realistic for high encounter frequency wave conditions. In contrast, Son and Nomoto (1982) and Araki et al. (2010) showed a large variation of maneuvering coefficients in following waves in which the encounter frequency is very low. Recently, CFD has been extended for free-running simulations, including quantitative agreement with experimental fluid dynamic (EFD) ship maneuvers in calm water (Stern et al., 2011) and waves (Sadat-Hosseini et al., 2011).

System identification techniques are developed in control engineering to build mathematical models for dynamical systems often using maneuvering coefficients based on experimental data. The least square (LS) is the one of the simplest and the extended Kalman filtering (EKF) (Lewis, 1986) is one of the most widely used methods in engineering. For ship hydrodynamics applications, the advantage of system identification is that all the maneuvering coefficients can be estimated by one or a few free-running trials as opposed to numerous captive model tests. Nonaka et al. (1972) employed LS to estimate maneuvering coefficients from EFD free-running data with random rudder motions and the Abkowitz mathematical model. However the estimated maneuvering coefficients were not accurate (i.e., large differences from those from the captive model tests), which is due to some of the derivatives drifting to the wrong values known as the simultaneous drift problem (Kang et al., 1984). Since some maneuvering coefficient signs or rough values are known from physical reasoning or previous research, the constrained least square (CLS) method using the generalized reduced gradient algorithm as developed by Lasdon et al. (1978) may be useful for estimating the maneuvering coefficients. The generalized reduced gradient algorithm is one of the optimization algorithms that make it possible to set general constraints on the optimization. The constraints could help avoiding the simultaneous drift problem.

Abkowitz (1980) employed EKF using full-scale trial data and the Abkowitz mathematical model. The 10/10 zigzag trial data was used to estimate the linear maneuvering coefficients and the $\delta = 35^\circ$ turning circle trial data was used for the nonlinear coefficients. The simulation using estimated maneuvering coefficients showed reasonable agreement with not only the 10/10 zigzag and $\delta = 35^\circ$ turning circle trial data, but also 20/20 zigzag trial result which were not used for the system identification.

Rhee and Kim (1999) employed EKF for SB free-running trial data (zigzag, turning circle, large angle zigzag tests, etc.) and the MMG mathematical model to find best trial type for system identification. The maneuvering coefficients reconstructed from the large angle zigzag test showed the smallest error with the original coefficients. Zhang and Zou (2011) employed support vector machine, one of the artificial intelligence methods, for SB free-running trial data (zigzag test) and the Abkowitz mathematical model for which the reconstructed coefficients showed close agreement with the original maneuvering coefficients. Several other researchers (for instance Shi et al., 2009) have employed EKF to estimate ship maneuvering coefficients. Tiano and Blanke (1997) used a simulated annealing search method, Zhou (1987) used recursive prediction error technique and Blanke and Knudsen (1998) showed the sensitivity approach is efficient in determining which parameter could be identified with satisfactory accuracy.

CFD free-running trial data has yet to be used for the system identification. Total and component hydrodynamic forces/moments acting on the ship/appendages/propeller can be estimated by CFD simulations while it is unknown for ship and difficult for appendages/propeller in experimental free-running measurements. Moreover it could be possible to compute reliable SB maneuvering simulations without any captive or free-running model experiments, but rather using system identification with CFD free-running trial data which offers significant benefits for ship design. CFD also has the capability of performing simulations at full-scale Reynolds numbers and environmental conditions (Bhushan et al., 2009).

Herein, maneuvering coefficients for a 4DOF SB method based on the MMG mathematical are estimated initially from captive tests (original maneuvering coefficients) and subsequently from CFD and EFD free-running trial data for calm water maneuvers using system identification to evaluate the mathematical model and determine the optimum approach for estimating the maneuvering coefficients. SB using the original maneuvering coefficients and CFD are validated using EFD for turning circle and zigzag maneuvers. EKF and CLS system identification are used with the SB, EFD and CFD free-running trial data to obtain new maneuvering coefficients, which are compared with the original coefficients. The new coefficients are then used with the SB method to predict the original and blind trajectories showing that CFD provides the optimum approach for estimating the maneuvering coefficients.

2. EFD method

All EFD free-running data was acquired in IIHR wave basin facility (Fig. 1). The wave basin has dimensions of $40 \times 20\ \mathrm{m}^2$ with 3 m water depth and is designed to test captive or radio-controlled model scale ships. There are six plunger-type wave makers on the east side of the basin to generate regular or irregular waves. The x-direction (length) main-carriage and y-direction (width) sub-carriage with yaw-direction turntable (heading) track or tow the model. The sub-carriage supports the model launch system and local flow measurement system traverse. The model launch system enables specification and replication of the free-running trial initial conditions.

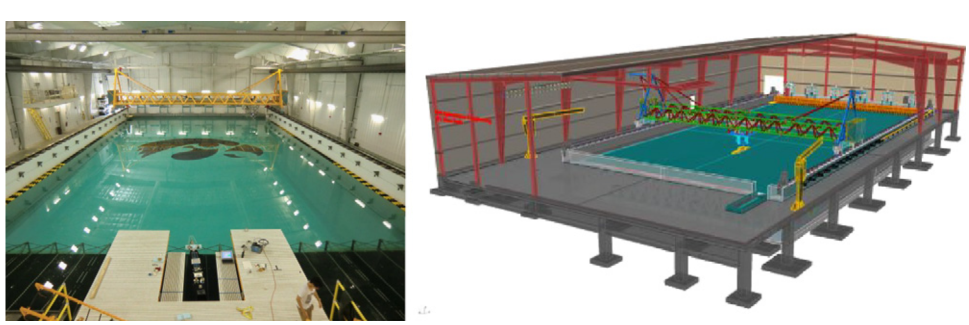

Fig. 1. Photograph and drawing of IIHR wave basin.

The 1/49 scaled model of ONR tumblehome (ONRT) appended with skeg, bilge keels, rudders, shafts and propellers with propeller shaft brackets was used for the experiments (Figs. 2 and 3 and Table 1). Roll, pitch and yaw rates of the model ship were measured by a fiber optical gyroscope (FOG) with 20 Hz. The roll, pitch and yaw angles are computed by integrating the measured rates. The accuracy of roll and pitch angles was $\pm 0.5^\circ$, while that for yaw angle was $\pm 1.0^\circ$ without considering the error due to drift. However the drift effect may be negligible because the gyro's yaw angle was reset through wireless communication just before the each trial is started.

The number of propeller revolutions (RPM) was counted by a photo micro sensor and the rudder angle was estimated from the number of pulse signals transmitted to the rudder-driving stepping motor. All the data of the free-running system was recorded with a frequency of 20 Hz. Meanwhile, the plane trajectory (x, y, ψ) of the model was recorded with a frequency of 10 Hz by the tracking system, which uses two-camera vision. The tracking cameras capture two LED lights placed on the deck of model.

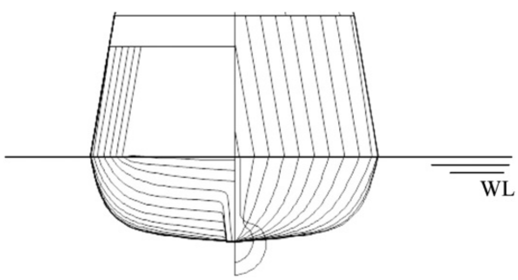

Fig. 2. Body plan of the ONRT model.

The mid-point of two LEDs corresponds to the longitudinal center of gravity (LCG) of the model, but 0.250 m higher than the vertical CG point due to space restriction. The model's horizontal position can be estimated indirectly by monitoring the rotational speed of each direction-motor. The maximum tracking speeds in the longitudinal x and transverse y directions are 2.5 m/s, while for the rotational speed it is 60°/s. Although the free-running and tracking systems are fully independent, all data can be synchronized receiving the start-signal of radio control.

With an onboard FOG and the carriage model tracking system enables a 5DOF measurement of a free-running model including xy-position, roll angle, pitch angle and yaw angle. In order to increase the reliability and accuracy of the 5DOF measurement and to enable measurement for all 6DOF of the free-running model, i.e. the heave motion, a 6DOF visual motion capture system (6DOF-VMCS) was developed and added to the tracking system. As shown in Fig. 4, a CCD camera was installed on the turntable of the sub-carriage that focuses on a calibration target plate attached on the deck of the free-running model. Digital images of the calibration target plate were acquired at a frame rate of up to 30 fps. The target images were analyzed with OpenCV library to determine the 6DOF motion of the free-running model relative to the carriage and sub-carriage/turntable. The 6DOF-VMCS measures displacements of x, y, z, ϕ, θ, ψ and compute their rates with numerical differentiation. Comparative tests show that the angles and displacements obtained with 6DOF-VMCS have good agreement with those obtained with the FOG and the carriage model tracking system.

The experimental procedure was as follows. First the model ship was fixed on the launch system by electromagnetics while the heave, roll and pitch are free. After the propeller starts to rotate, the model was accelerated by the launch system to the

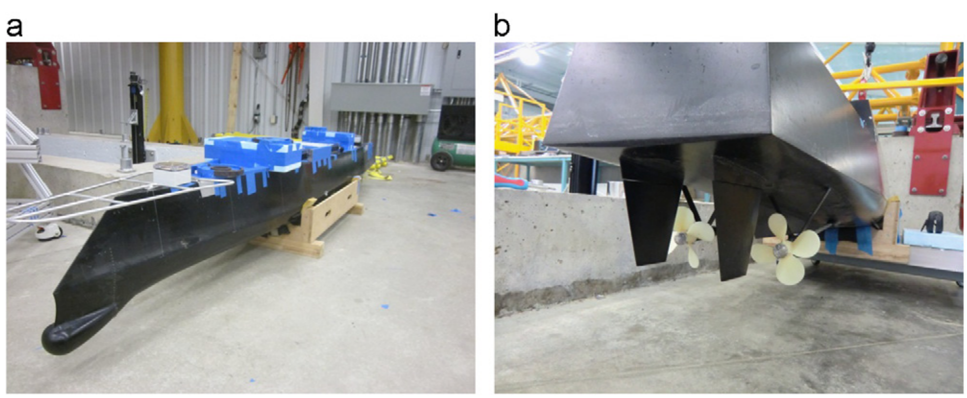

Fig. 3. Bow and stern of the ONRT model: (a) bow; (b) stern.

Table 1
Principal particulars of the ONRT vessel.

	Model scale
Length: L	3.147 m
Breadth: B	0.384 m
Depth: D	0.266 m
Draft: d	0.112 m
Displacement: W	72.6 kg
Metacentric height: GM	0.0424 m
Natural roll period: T_ϕ	1.644 s
Rudder area: A_R	$0.012\ m^2 \times 2$
Block coefficient: C_b	0.535
Vertical position of CoG from waterline (downward positive): OG	$-0.392 \times d$
Radius of gyration in pitch: κ_{yy}	$0.25 \times L$
Maximum rudder angle: δ_{max}	$\pm 35^\circ$

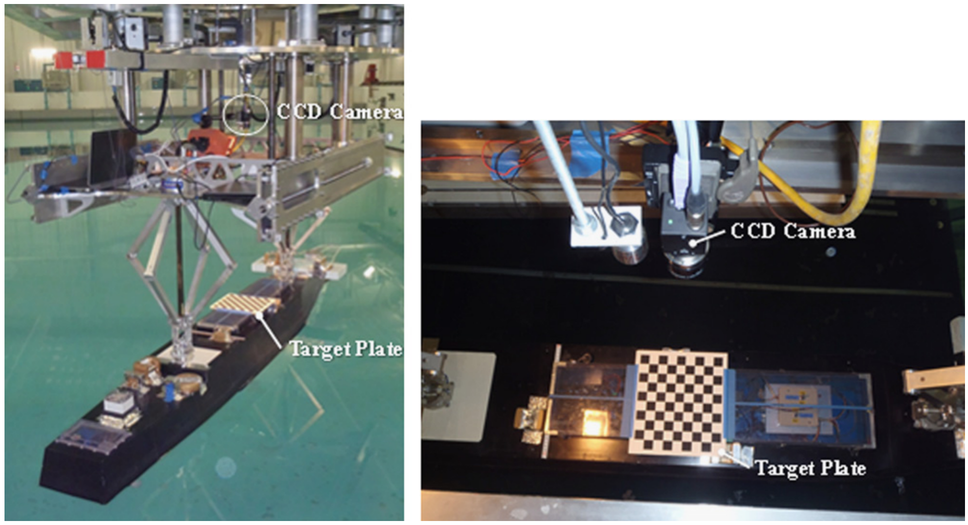

Fig. 4. 6DOF visual motion capture system.

target speed and tows the model with constant speed for a specified period and then releases it. After the model was released from the launch system, the model ran straight without any rudder deflection for a specified period and started maneuvering after pre-set time. The propeller RPM was kept constant while running.

A detailed description of the wave basin and wavemakers, carriage model tracking, 6DOF visual motion capture and free-running 6DOF systems, model geometry and ballasting, and free-running trials tests in calm water and waves is provided by Sanada et al. (2012).

3. System-based method

Ship motions can be described with a 6DOF mathematical model as shown in Eq. (1). The mass distribution is symmetric with respect to the ship center plane, thus the y_{CG}, I_{xy} and I_{yz} are zero. Additionally I_{xz} is neglected as small for ships with approximate fore and aft symmetry.

$$\begin{aligned}
&m(\dot{u}-vr+wq)=X\\
&m(\dot{v}-wp+ur)=Y\\
&m(\dot{w}-uq+vp)=Y\\
&ptI_x\dot{p}\rightarrow+(I_z-I_y)qr=K\\
&I_x\dot{p}+(I_z-I_y)qr=K\\
&I_z\dot{r}+(I_y-I_x)pq=N
\end{aligned}\tag{1}$$

In calm water, if the order of the surge motion is assumed to be first order, the sway, yaw motion would be second order because the forward velocity is much larger than the sway or yaw velocities. The heave, roll and pitch motions would be third order. Therefore ship motions in calm water can be simplified to 3DOF motions (surge-sway-yaw). However in practice roll motion can be observed during hard turning in calm water because of the rudder and centrifugal forces. Therefore, the roll motion is included and a 4DOF mathematical model (surge-sway-roll-yaw) is used, as shown in Eq. (2), which can be extended to wave cases. In wave, roll motion would be second order due to the wave-induced motion but heave and pitch motions are still third order because of the large restoring forces and moments. Here I_y is assumed to be the same as I_z because the ship is a slender body.

$$\begin{aligned}
&m(\dot{u}-vr)=X\\
&m(\dot{v}+ur)=Y\\
&I_x\dot{p}=K\\
&I_x\dot{p}=K
\end{aligned}\tag{2}$$

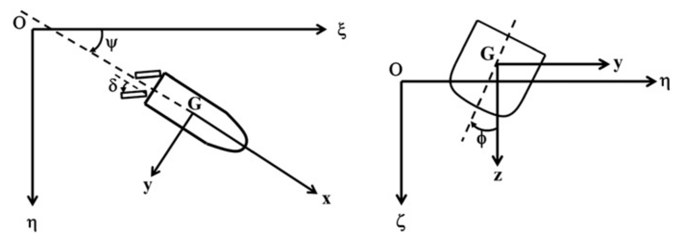

Fig. 5. Coordinate system for 4DOF SB model.

Therefore a 4DOF MMG mathematical model is used for the SB simulations as shown in Eqs. (3)–(6). Horizontal body axes (Hamamoto and Kim, 1993) and steadily translating axes which move with a wave, shown in Fig. 5, are used for the coordinate system. Fujii's rudder model (Fujii and Tuda, 1961), shown in Eqs. (8)–(25), is used for rudder forces/moments.

$$(m+m_x)\dot{u}-(m+m_y)vr=T(u;n)-R(u)+X_{rr}(u)r^2+X_{vr}(u)vr +X_{vv}(u)v^2+X_R(\delta,u,v,r)\tag{3}$$

$$(m+m_y)\dot{v}+(m+m_x)ur=Y_r(u)r+Y_v(u)v+Y_\phi(u)\phi+Y_{rrr}(u)r^3 +Y_{rrv}(u)r^2v+Y_{rvv}(u)rv^2+Y_{vvv}(u)v^3+Y_R(\delta,\phi,u,v,r)\tag{4}$$

$$(I_x+J_x)\dot{p}=m_xz_Hur+K_r(u)r+K_p(u)p+K_v(u)v+K_\phi(u)\phi-mgGZ(\phi) +K_{rrr}(u)r^3+K_{rrv}(u)r^2v+K_{rvv}(u)rv^2+K_{vvv}(u)v^3+K_R(\delta,u,v,r)\tag{5}$$

$$(I_z+J_z)\dot{r}=N_r(u)r+N_v(u)v+N_\phi(u)\phi+N_{rrr}(u(u))r^3+N_{rrv}(u)r^2v +N_{rvv}(u)rv^2+N_{vvv}(u)v^3+N_R(\delta,\phi,u,v,r)\tag{6}$$

here

$$\begin{bmatrix}K_r & K_v & K_{rrr} & K_{rrv} & K_{rvv} & K_{vvv}\end{bmatrix}^T=z_H\begin{bmatrix}Y_r & Y_v & Y_{rrr} & Y_{rrv} & Y_{rvv} & Y_{vvv}\end{bmatrix}^T\tag{7}$$

$$X_R=-(1-t_R)F_N\sin\delta\tag{8}$$

$$Y_R=-(1+a_H)F_N\cos\delta\cdot\cos\phi\tag{9}$$

$$K_R=z_{HR}(1+a_H)F_N\cos\delta\tag{10}$$

$$N_R=-(x_R+a_Hx_H)F_N\cos\delta\cdot\cos\phi\tag{11}$$

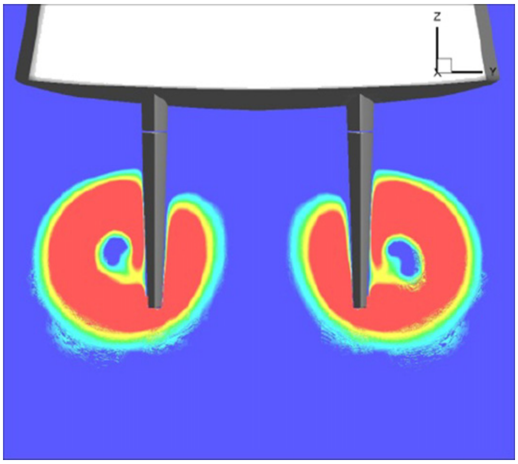

Fig. 6. CFD propeller race around the rudders.

$$F_N = \frac{1}{2}\rho A_R f_\alpha U_R^2 \sin\alpha_R \tag{12}$$

The propeller race is only effective on about 2/3 of the rudder area as shown in Fig. 6. The last 1/3 of the rudder generates lift by the ship wake. Therefore the effective rudder inflow speed should be described by Eqs. (13)–(23) (Kose et al., 1981).

$$U_R = \sqrt{u_R^2 + v_R^2} \tag{13}$$

$$u_R \cong \sqrt{\frac{A_{R1}}{A_R}u_{Rp}^2 + \frac{A_{R2}}{A_R}u_{R0}^2} = \varepsilon u_p\sqrt{\eta\left\{1+\kappa\left(\sqrt{1+\frac{8K_T}{\pi J^2}}-1\right)\right\}^2 + (1-\eta)} \tag{14}$$

$$u_{R0} = (1-w_R)u \tag{15}$$

$$u_{Rp} = u_{R0} + 0.6(u_\infty - u_p) = u_p\left\{\varepsilon + 0.6\left(\frac{u_\infty}{u_p} - 1\right)\right\} \tag{16}$$

$$u_\infty = u_p\sqrt{1+\frac{8K_T}{\pi J^2}} \tag{17}$$

$$u_p = (1-w_p)u \tag{18}$$

$$\varepsilon = \frac{1-w_R}{1-w_p} \tag{19}$$

$$\kappa = 0.6/\varepsilon \tag{20}$$

$$\frac{A_{R1}}{A_R} = \eta = 2/3 \tag{21}$$

$$\frac{A_{R1}}{A_R} = 1-\eta = 1/3 \tag{22}$$

$$v_R = -\gamma_R(v + l_R r) \tag{23}$$

$$f_\alpha = \frac{6.13\Lambda_R}{\Lambda_R + 2.25} \tag{24}$$

$$\alpha_R = \delta - \gamma_R\frac{v_R}{u_R} \tag{25}$$

In the mathematical model, resistance is estimated from a captive model experiment and the thrust is estimated from propeller open water tests in calm water as described in Umeda et al. (2008). Roll restoring moment, *mgGZ*, is estimated from hydrostatic calculations in calm water. Maneuvering coefficients including heel-induced hydrodynamic derivatives are estimated from calm water captive model experiments (Hashimoto et al., 2008). Roll damping is estimated from roll decay model tests (Umeda et al., 2008). The values of all the original hydrodynamic and rudder maneuvering coefficients used in the SB mathematical model are listed in Table 2. Here, although the ship model has twin-propeller/twin-rudder, empirical rudder-related coefficients for single-propeller/single-rudder model are used for this SB propeller/rudder model. However it is acceptable since Lee et al. (1988) showed using single-propeller/single-rudder coefficients for twin-propeller/twin-rudder ship has little influence to predict ship maneuvers in practice. For the ONRT, flow-straightening effect coefficients γ_R, shown in Eq. (23), are adjusted to 0.70 because the rudders are situated far from the hull compared to other conventional ships which could weaken the flow-straightening effect. Thus the flow-straightening effect caused by the hull would be smaller, i.e. the effect of sway and yaw motion on the rudders would be larger. Also $-1.0L$ is used for the correction coefficient for sway l_R shown in Eq. (23). The values of γ_R and l_R are empirically developed from other model experiments (Kose et al., 1981). The empirical values are also used for the interaction force coefficients induced on the hull by rudder nominal force t_R, a_H, z_{HR} and x_H as shown in Eqs. (8)–(11). All maneuvering coefficients values related to rudder force are listed in Table 2.

Table 2
Values of maneuvering and rudder coefficients used in 4-DoF nonlinear SB model.

Coefficient	Value	Source
ε	1.0	Empirical (Kose et al., 1981)
γ_R	0.70	Empirical (Kose et al., 1981)
l_R/L	-1.00	Empirical (Kose et al., 1981)
t_R	0.30	Empirical (Kose et al., 1981)
a_H	0.25	Empirical (Kose et al., 1981)
z_{HR}/d	0.854	Geometrical (center of rudder)
x_H/L	-0.45	Empirical (Kose et al., 1981)
m_x	0.01307	Empirical (Motora, 1960)
X_{vv}	-0.08577	Static drift test (Umeda et al., 2008)
X_{vr}	0.052212	CMT (Umeda et al., 2008)
X_{rr}	-0.02126	CMT (Umeda et al., 2008)
m_y	0.10901	Strip theory
Y_v	-0.30015	Static drift test (Umeda et al., 2008)
Y_r	-0.08316	CMT (Umeda et al., 2008)
Y_{vvv}	-1.77272	Static drift test (Umeda et al., 2008)
Y_{vvr}	0.261986	CMT (Umeda et al., 2008)
Y_{vrr}	-0.79966	CMT (Umeda et al., 2008)
Y_{rrr}	0.173916	CMT (Umeda et al., 2008)
Y_ϕ	-0.00051	Static heel test (Umeda et al., 2008)
J_{xx}	4.1E$-$05	Free roll decay test (Umeda et al., 2008)
z_H	0.852	Static drift test (Umeda et al., 2008)
K_p	-0.2429	Free roll decay test (Umeda et al., 2008)
K_ϕ	0.000626	Static heel test (Umeda et al., 2008)
J_{zz}	0.00789	Strip theory
N_v	-0.09323	Static drift test (Umeda et al., 2008)
N_r	-0.05494	CMT (Umeda et al., 2008)
N_{vvv}	-0.53235	Static drift test (Umeda et al., 2008)
N_{vvr}	-0.62895	CMT (Umeda et al., 2008)
N_{vrr}	-0.13897	CMT (Umeda et al., 2008)
N_{rrr}	-0.00446	CMT (Umeda et al., 2008)
N_ϕ	-0.00511	Static heel test (Umeda et al., 2008)

Eqs. (26) and (27) are used to compute the thrust. The value of the thrust reduction t_p and the effective propeller wake fraction w_p, shown in Eqs. (26) and (27), are assumed to be zero because of the large space between the ship hull and the propellers as shown in Fig. 3b.

$$T = (1-t_p)\rho n^2 D_p^4 K_T(J)_R \tag{26}$$

$$J = \frac{(1-w_p)u}{nD_p} \tag{27}$$

4. CFD method

The code CFDShip-Iowa v4 (Carrica et al., 2010) is used for the CFD computations. The CFDShip-Iowa is an overset, block structured CFD solver designed for ship applications using either absolute or relative inertial non-orthogonal curvilinear coordinate system for arbitrary moving but non-deforming control volumes. Turbulence models include blended $k-\varepsilon/k-\omega$ based isotropic and anisotropic RANS, and DES approaches with near-wall or wall functions. A single-phase level-set method is used for free-surface capturing. Captive, semi-captive, and full 6DOF capabilities for multi-objects with parent/child hierarchy are available. The actual propeller or body propeller model can be employed for propulsion. The PID controllers are available to correct the error between a measured process variable and a desired setpoint by calculating and then outputting a corrective action that can adjust the process accordingly. Numerical methods include advanced iterative solvers, higher order finite differences with conservative formulation, PISO or projection methods for pressure–velocity coupling, and parallelization with MPI-based domain decomposition. Dynamic overset grids use SUGGAR to compute the domain connectivity.

For the current simulations, absolute inertial earth-fixed coordinates are employed with $k-\varepsilon/k-\omega$ turbulence model using no wall function. The location of the free-surface is given by the 'zero' value of the level-set function, positive in water and negative in air. The 6DOF rigid body equations of motion are solved to calculate ship linear and angular motions. A simplified body force model is used for the propeller which prescribes axisymmetric body force with axial and tangential components. The propeller model requires the open water curves and advance coefficients as input, and provides the torque and thrust forces. The open water curves are defined as a second order polynomial fit of the experimental KT(J) and KQ(J) curves. The advance coefficient is computed using Eq. (27) with neglecting the wake effects. Also, the thrust force is calculated using Eq. (26) without using tp factor. Herein, two PID controllers are used. The heading controller acting on the rudders are responsible to turn the rudders to keep the ship in the desired direction. The speed controller acting on the body force propeller model is responsible to rotate the propellers at appropriate RPM to keep the ship at the desired speed. The heading controller uses P=1 for the proportional gain and zero for both the integral and derivative gains mimicking EFD setup.

The governing equations are discretized using finite difference schemes on body-fitted curvilinear grids. The time derivatives in the turbulence and momentum equations are discretized using 2nd order finite Euler backward difference. Convection terms in the turbulence and momentum equations are discretized with higher order upwind formula. The viscous term in momentum and turbulent equations are computed with similar considerations using a second order difference scheme. The PISO algorithm is solved using the PETSc toolkit. SUGGAR is used to obtain the overset interpolation information. A MPI-based domain decomposition approach is used, where each decomposed block is mapped to one processor.

In order to achieve a large size computational domain with reasonable number of grid points, a cylinder shape computational domain was used. The radius of domain is 4.5 times larger than ship length and the domain extends from $z=-1$ L to $z=0.25$ L in vertical direction. The ship axis is aligned with x axis, with the bow at $x=0$ and the stern at $x=1$. The y axis is positive to starboard with z pointing upward. The free surface at rest lies at $z=0$. The free model is appended with skeg, bilge keels, superstructure, rudders, rudder roots, shafts, and propeller brackets mimicking the EFD conditions but not appended with actual propellers.

The computational grids are overset, with independent grids for the hull, superstructure, appendages, refinement and background, and then assembled together to generate the total grid. The boundary layer grids are generated with a hyperbolic grid generator using a double-O topology, one each for the starboard and the port sides. Grid spacing at the hull is designed to yield $y+<1$ for a wide range of Froude numbers. The bilge keels and skeg use H topology grids and overset the boundary layer grids. The superstructure grid oversets the boundary layer grids and is constructed with wrapped face topology. Two double-O grids are used for each rudder such that inboard and outboard sides are conformal. O topology grids are used for propeller shafts and struts. A Cartesian grid is used to impose the far-field boundary conditions and to resolve the flow far from the hull, with a refinement Cartesian block closer to the ship. The total number of grid points is 12.1 M for free model simulations. Details of the grids are shown in Table 3 and Fig. 7.

For boundary conditions, the no-slip condition is applied on the solid surfaces. The far field boundary conditions are imposed on the top and bottom of background. The inlet boundary condition with zero pressure gradient is used for the surrounding surface of large cylinder shape background.

Table 3
Grids for free model simulations.

Name	**Size (grid points)**	**# of procs**	**Type**
Hull S/P[a]	$199\times61\times104$ (1.26 M $\times$ 2)	12 ($\times$2)	Double O
Skeg S/P	$61\times49\times40$ (0.12 M $\times$ 2)	1 ($\times$2)	O
Bilge Keel S/P	$99\times45\times50$ (0.23 M $\times$ 2)	2 ($\times$2)	H
Rudder Root Collar S/P	$121\times35\times28$ (0.12 M $\times$ 2)	1 ($\times$2)	O
Rudder Root Gap S/P	$121\times51\times19$ (0.12 M $\times$ 2)	2 ($\times$2)	Conformal to Collar
Rudder Outer S/P	$61\times36\times55$ (0.12 M $\times$ 2)	1 ($\times$2)	Double O
Rudder Inner S/P	$61\times36\times55$ (0.12 M $\times$ 2)	1 ($\times$2)	Double O
Rudder Gap S/P	$121\times51\times19$ (0.12 M $\times$ 2)	2 ($\times$2)	Conformal to Inner and Outer
Shaft Collar S/P	$39\times50\times57$ (0.11 M $\times$ 2)	1 ($\times$2)	O
Shaft Proper S/P	$74\times41\times37$ (0.11 M $\times$ 2)	1 ($\times$2)	O
Shaft Tip S/P	$110\times117\times100$ (1.29 M $\times$ 2)	12 ($\times$2)	O with end pole
Strut Outer S/P	$69\times34\times50$ (0.12 M $\times$ 2)	1 ($\times$2)	O
Strut Inner S/P	$69\times34\times50$ (0.12 M $\times$ 2)	1 ($\times$2)	O
Superstructure	$165\times61\times85$ (0.86 M)	8	Wrap
Refinement	$145\times81\times113$ (1.33 M)	12	Cartesian
Background	$213\times84\times113$ (2.02 M)	20	O
Total	**(12.1 M)**	**116**	

[a] S/P: Starboard/Port

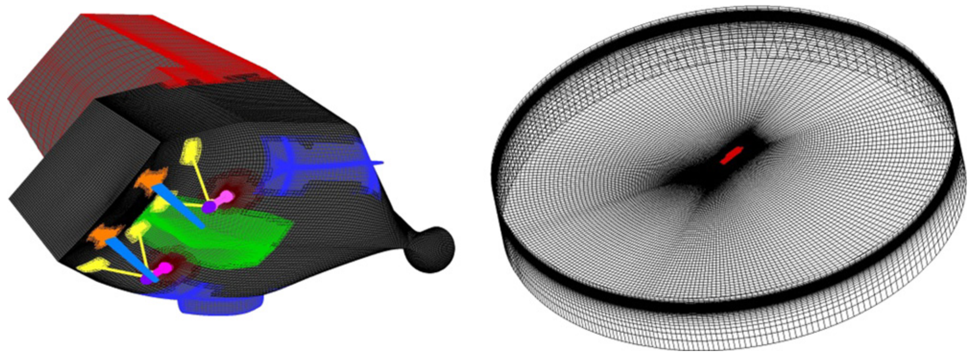

Fig. 7. CFD overset grids for ONRT hull, appendages and background.

Table 4
EFD, CFD and SB test matrices.

	Test	Fr	δ (deg)	ψ_c^* (deg)
EFD	Turning Circle	0.1, 0.2	25, 35	N/A
	Zigzag	0.1, 0.2	10, 20	10, 20
	Large Angle Zigzag	0.2	35	90
CFD and SB	Turning Circle	0.2	25	N/A
	Zigzag	0.2	20	20
	Large Angle Zigzag	0.2	35	90

ψ_c^*: target yaw angle

Table 5
Constraints for the coefficients.

$(X_{vr}+m_y)$, t_R, a_H, z_H, z_{HR}, x_H, m_x, m_y, J_{xx}, J_{zz}, γ_R, ε	≥ 0
X_{rr}, X_{vv}, Y_v, Y_{vvv}, (Y_r-m_x), K_p, N_r, N_{rrr} l_R,	≤ 0
t_R, γ_R,	≤ 1

The simulation test matrix is shown in Table 4. For all cases the RPM is fixed during the simulation mimicking EFD setup. The propeller RPS required to achieve the target ship speed is obtained by performing a self-propelled simulation starting from static ship conditions using speed controller.

The free-running verification studies have not been done yet. For captive tests, a verification study has been performed for a much simpler geometry of ONRT (only bilge keels as appendages) with the absence of moving propellers and rudders (Sadat-Hosseini et al., 2011). The verification study was carried out for the grid size of 3.48 M coarsened and refined by a factor of $\sqrt{2}$. The study was performed for 10° of heel angle for full Fr curve whereby the results were obtained for the full Fr range of interest in a single simulation. The grid verification was investigated not only for resistance and sinkage and trim but also for side force and roll and yaw moments which are significantly important in complicated cases such as broaching. The average of grid uncertainty for resistance and side forces, roll and yaw moments and sinkage and trim motions was U_g=2.75 in %DR (Dynamic Range) with the maximum grid uncertainty of about 6%DR for side force and yaw moment. The validation errors were reported for resistance, 10° and 20° static heel and 5°, 10°, 15° static drift tests. For resistance test, the average of validation errors for resistance force, sinkage and trim for full Fr curve was 2.88 in %D and 2.5% in %DR with largest error for sinkage (E=5%D). For 10° static heel test, the average of validation errors for resistance and side forces, roll and yaw moments and sinkage and trim motions for full Fr curve was 17.3 in %D and 9.9 in %DR. The largest errors were obtained for side force, yaw moment and trim motion with E~28%D. The same order of validation errors were found for the 20° static heel test. For 5°, 10° and 15° static drift tests, the average of validation errors of all cases for resistance and side forces, roll and yaw moments and sinkage and trim motions was 20.3%D and 1.4%DR for full Fr curve. The largest errors were found for roll moment and trim motion.

5. System identification

System identification is used with the SB mathematical model, shown in Eqs. (3)–(6), along with the EFD, SB and CFD free-running trial data to obtain new maneuvering coefficients. In previous studies (Abkowitz, 1980), EFD free-running state variables were used to evaluate both the right-hand and left-hand sides of the mathematical model. Empirical values were used for the added mass and added moment of inertia to estimate the maneuvering coefficients. Herein not only CFD state variables but also total forces/moments are used such that the added mass and added moment of inertia can be predicted without using empirical values. Furthermore, CFD propulsion forces, rudder forces and hydrostatic forces can be separated from the total hydrodynamic forces. This helps is evaluating the maneuvering coefficients and the empirical rudder coefficients effectively. Herein the extended EKF and CLS system identification methods are used. The EFD data are unified to 30 Hz; 6DOF-VMCS data (30 Hz) are used for state variables and rudder motions from the free-running system (20 Hz) are interpolated using the spline interpolation.

5.1. Extended Kalman filtering

The EKF method optimizes the target parameters based on the equations of motion and the specified state variables. Therefore it does not require any acceleration terms for input which are noisy in the EFD data. This is one of the largest advantages of the EKF. The EKF method is used with SB, EFD and CFD inputs to estimate the maneuvering coefficients. The details of EKF are provided in Appendix A.

5.2. Constrained least square method using generalized reduced gradient algorithm

CLS is employed to estimate the hydrodynamic and rudder maneuvering coefficients with SB and CFD inputs. The constraints are set as shown in Table 5 according to physical reasoning and previous research (Kose et al., 1981). As shown in Fig. 8, the searching starts from initial values and aims at a minimum point of the error with constraints/interruptions. Therefore the optimized solutions do not reach to the minimum point at times. Herein the thrust, rudder forces, hydro-static forces and the resistance for x force are excluded from the total forces. The details of the generalized reduced gradient algorithm are provided in Appendix B.

6. Comparison between EFD, CFD and SB

6.1. Speed test

For CFD, the required propellers RPS for each Froude number is predicted with 2DOF sinkage and trim simulations in calm water with a speed controller set at the desired Froude number. The resulting RPS is later prescribed in the free model simulations. This propeller model is the simplest that can be applied and has some limitations. The most important is that the thrust and torque do not depend on the local flow field near the propellers, but on the average velocity of the ship. In addition, the body force is axisymmetric and side forces are neglected. Restrictions on the ability of the propeller model to handle strongly unsteady flows can also be expected. For SB, RPS is predicted from the open water curves similar to CFD. Also, the wake effect and side forces are neglected. Fig. 9 shows the results of EFD, CFD and SB speed tests. The error between SB and EFD is 11% at $Fr=0.20$ and the error between CFD and EFD is 14%. These discrepancies are likely due to the non-interactive nature of the propeller model that neglects the boundary layer and thus results in higher advance velocities and higher rotational speeds to achieve the same thrust.

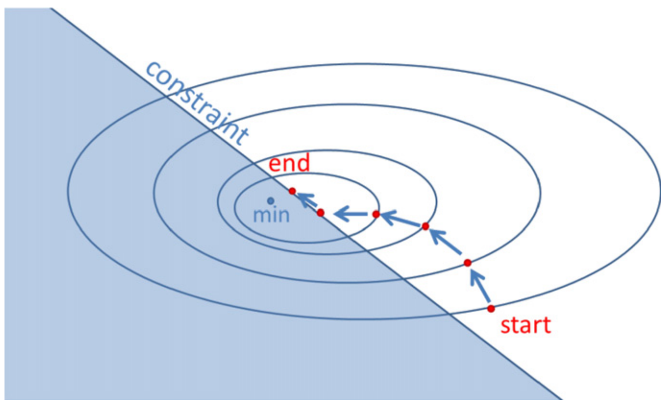

Fig. 8. Idea of the CLSM-GRG algorithm.

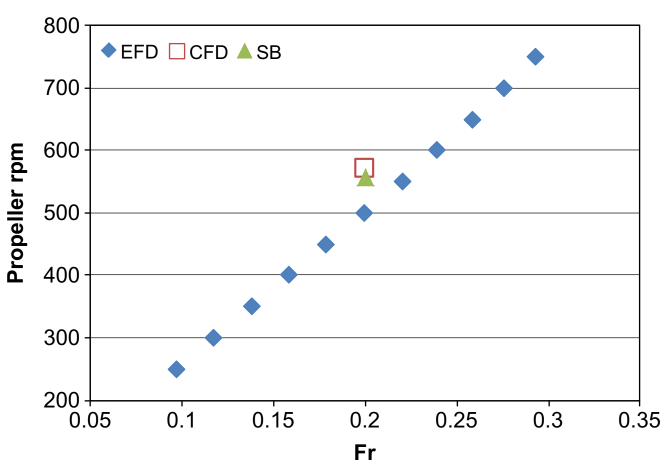

Fig. 9. EFD, CFD and SB comparison of propeller RPM in speed test.

6.2. Turning circle, zigzag and large Angle zigzag Tests

The EFD, CFD and SB maneuvering test conditions are listed in Table 4. The large angle zigzag test is a combination of turning circle and zigzag tests. Hereafter, turning circle test (Fr=0.20, $\delta=25^\circ$) is entitled as T25, zigzag test (Fr=0.20, $\delta/\psi_c=20/20$) as Z20, and large angle zigzag test (Fr=0.20, $\delta/\psi_c=35/90$) as Z90. Trajectories of T25, Z20, and Z90 are shown in Fig. 10. EFD trajectories are the average of 5 trials. In Fig. 10, the error bars of EFD trajectories are fairly small which show the good reproducibility of the EFD tests. Hereafter typical cases are chosen for each EFD data.

Trajectories of T25 are shown in Fig. 10a. The errors of CFD and SB turning circle simulations and EFD results are listed in Table 6. Here the values in Table 2 are used for the SB maneuvering coefficients. The steady state variables errors (u E%D, v E%D, r E%D, ϕ E%D) are computed by averaging each state variable (u, v, r, ϕ) after model motion converges to a steady state. "SSV Av. E%D" in Table 6 indicates the average of all steady state variables errors. Turning geometries (A, T, R), shown in Table 6, are computed based on the American Bureau of Shipping's maneuvering guide (2006). Here "A E%D" indicates error of advance distance, "T E%D" for transfer distance, "R E%D" for turning radius, and "ATR Av. E%D" is the average of all turning geometry errors. And "Total Av. E%D" is computed by averaging "SSV Av. E%D" and "ATR Av. E%D". The CFD trajectory shows close agreement with EFD while SB shows small error. The time series of the turning circle tests are shown in Fig. 11. The CFD errors are smaller than those of SB in general. Since the roll angle values are small (approximately -3° to -2°), the error of the roll angle seems to be larger than the

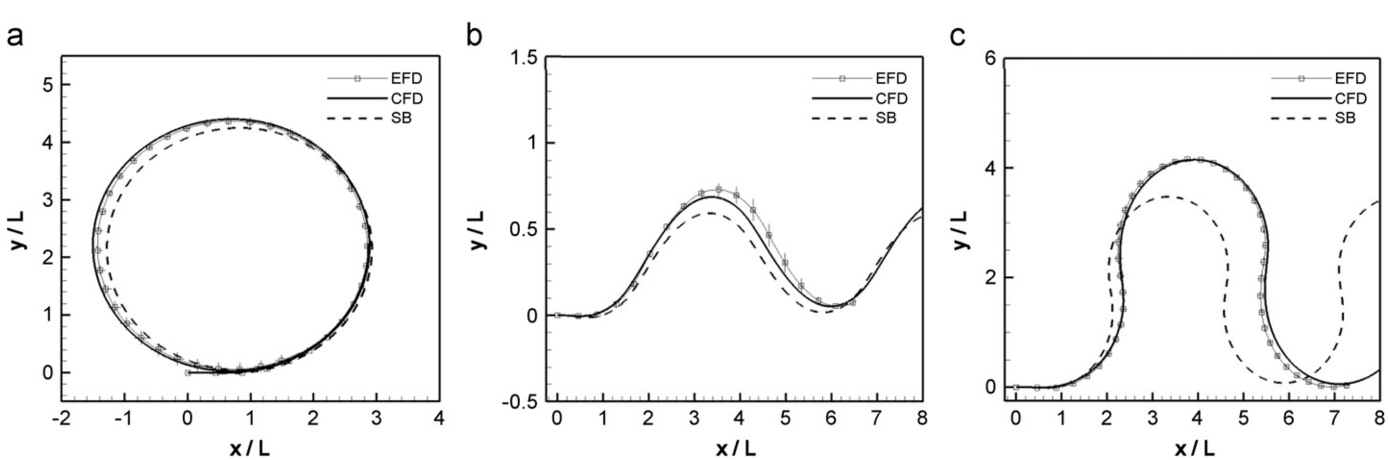

Fig. 10. Trajectories of free-running simulations: (a) Turning circle ($\delta=25^\circ$, Fr=0.20); (b) zigzag ($\delta/\psi_c=20/20$, Fr=0.20); (c) large angle zigzag ($\delta/\psi_c=35/90$, Fr=0.20).

Table 6
CFD and SB turning circle simulation errors from the EFD results (Fr=0.20, $\delta=25^{\circ}$).

	u E%D	v E%D	r E%D	ϕ E%D	SSV Av. E%D	A E%D	T E%D	R E%D	ATR Av. E%D	Total Av. E%D
CFD	−1.68	5.93	−3.59	17.5	7.18	−1.23	−2.19	−1.34	1.59	**4.38**
SB	−2.62	−12.8	−5.95	20.5	10.5	−2.21	4.06	2.44	2.90	**6.70**

Table 7
CFD and SB zigzag simulation errors from the EFD results (Fr=0.20, $\delta/\psi_c=20/20$).

	K20 E%D	T20 E%D	N20 E%D	KTN Av. E%D	1st OS E%D	2nd OS E%D	OS Av. E%D	Total Av. E%D
CFD	0.88	−2.67	5.90	3.15	−1.54	−2.52	2.03	**2.29**
SB	−3.64	−45.7	−3.65	17.7	−6.92	6.30	6.61	**12.2**

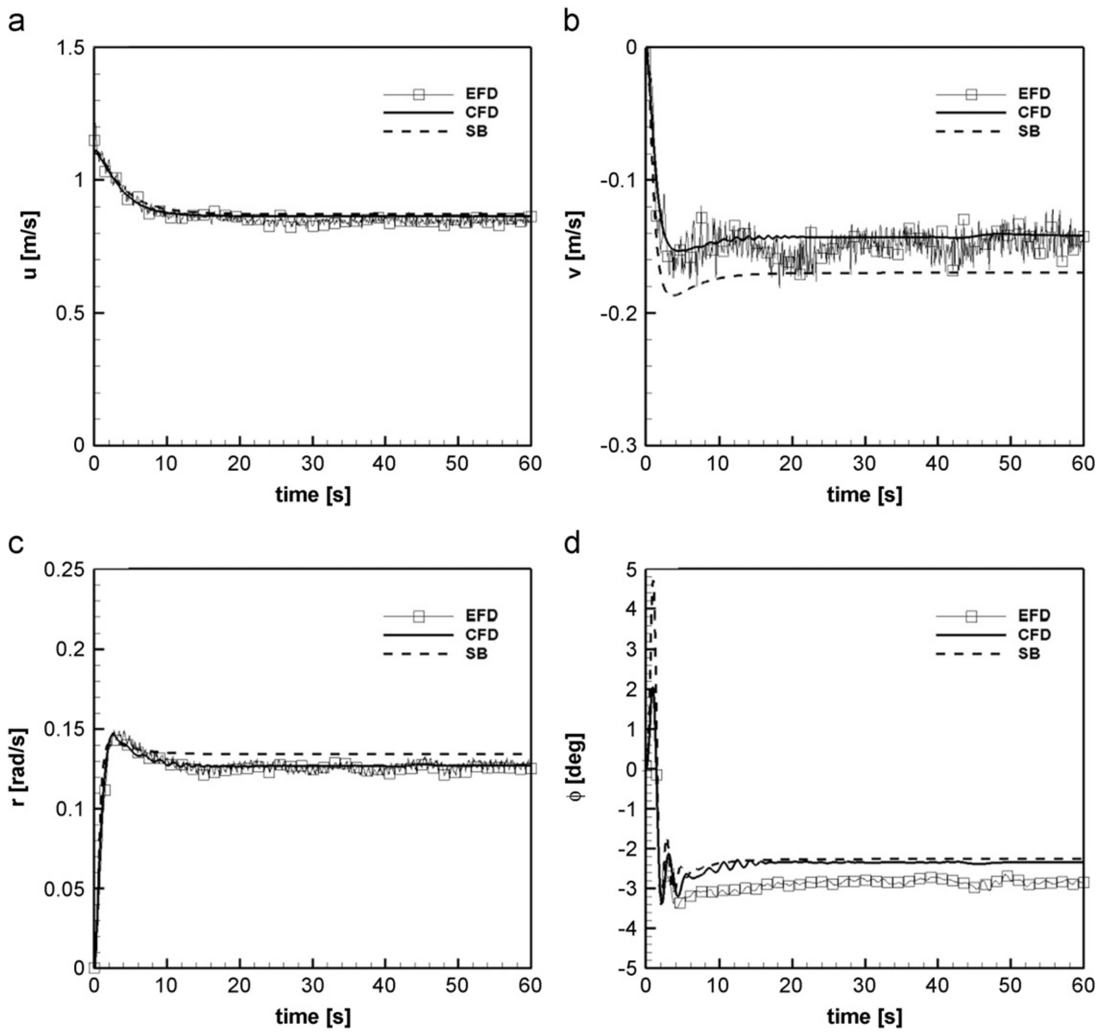

Fig. 11. Time series of turning circle test ($\delta=25^{\circ}$, Fr=0.20): (a) surge velocity; (b) sway velocity; (c) yaw rate; (d) roll angle.

other state variables. Moreover these small roll angles are not effective to the maneuvers.

The errors of CFD and SB's Z20 and EFD result are listed in Table 7. From zigzag test, the ship maneuverability can be defined by steering quality indices K_s, T_s and N_s (Norrbin, 1963). K_s, T_s and, N_s are estimated by fitting the time series to nonlinear 1st order Nomoto model Eq. (28) using least square method. In Table 7, the errors of K_s, T_s and N_s are shown as "K20 E%D", "T20 E%D" and "N20 E%D". The average of these errors is "KTN Av. E%D". The errors of 1st and 2nd overshoot angles are "1st OS E%D" and "2nd OS E%D" and the average of these overshoot angles are "OS Av. E%D". "Total Av. E%D" is the average of "KTN Av. E%D" and "OS Av. E%D". In Table 7, CFD shows small errors while SB shows larger errors.

$$T_s\dot{r}+N_s r^3+r=K_s\delta \tag{28}$$

Fig. 10b shows trajectories of the Z20. In Fig. 10b, the CFD trajectory shows good agreement with the EFD while the SB

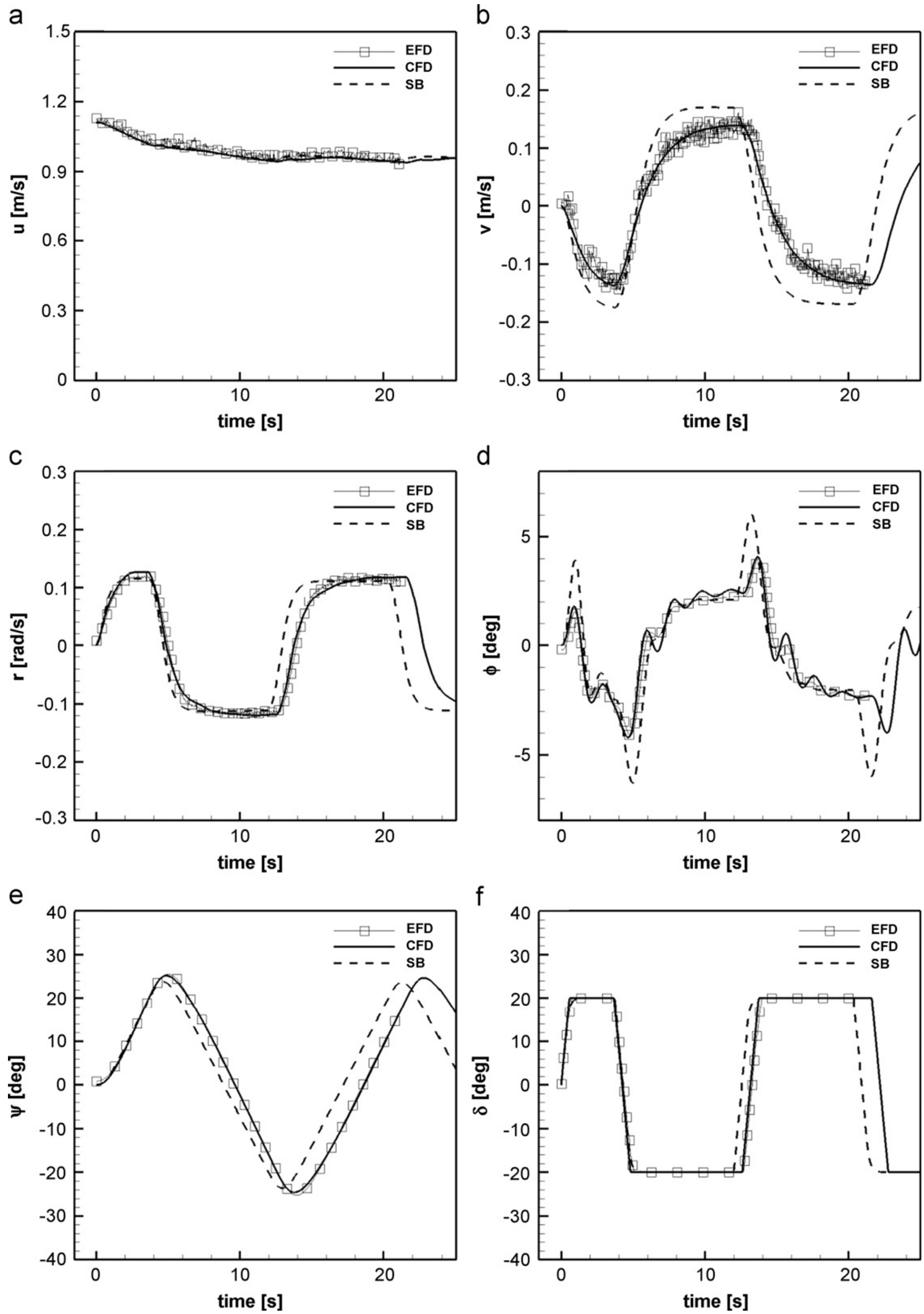

Fig. 12. Time series of zigzag test (δ/ψ_c=20/20, Fr=0.20): (a) surge velocity; (b) sway velocity; (c) yaw rate; (d) roll angle; (e) yaw angle; (d) rudder angle.

trajectory shows smaller turning than that of EFD and CFD. The time series of the EFD, CFD and SB zigzag test are shown in Fig. 12. In surge velocity (Fig. 12a), CFD and SB show good agreement with that of EFD. Similarly CFD shows good agreement for the other state variables. However SB over predicts the sway velocity. In roll angle (Fig. 12d), SB over predicts the peak of the roll motion just after the rudder deflection because the SB's roll equation Eq.(5) does not include the coupling terms related to sway acceleration or yaw angular acceleration which are not negligible just after large rudder deflection.

The errors of CFD and SB's Z90 and EFD result are listed in Table 8. The criteria of the errors are the same as the aforesaid zigzag tests. Similarly to the Z20 zigzag simulations, CFD shows smaller errors with EFD results than SB in Table 8. Especially SB's

Table 8
CFD and SB large angle zigzag simulation errors from the EFD results (Fr=0.20, δ/ψ_c=35/90).

	K90 E%D	T90 E%D	N90 E%D	KTN Av. E%D	1st OS E%D	2nd OS E%D	OS Av. E%D	Total Av. E%D
CFD	4.32	−20.7	−1.61	8.88	−1.23	−1.60	1.42	**5.15**
SB	25.1	−47.7	100	57.6	−0.59	−0.72	0.65	**29.1**

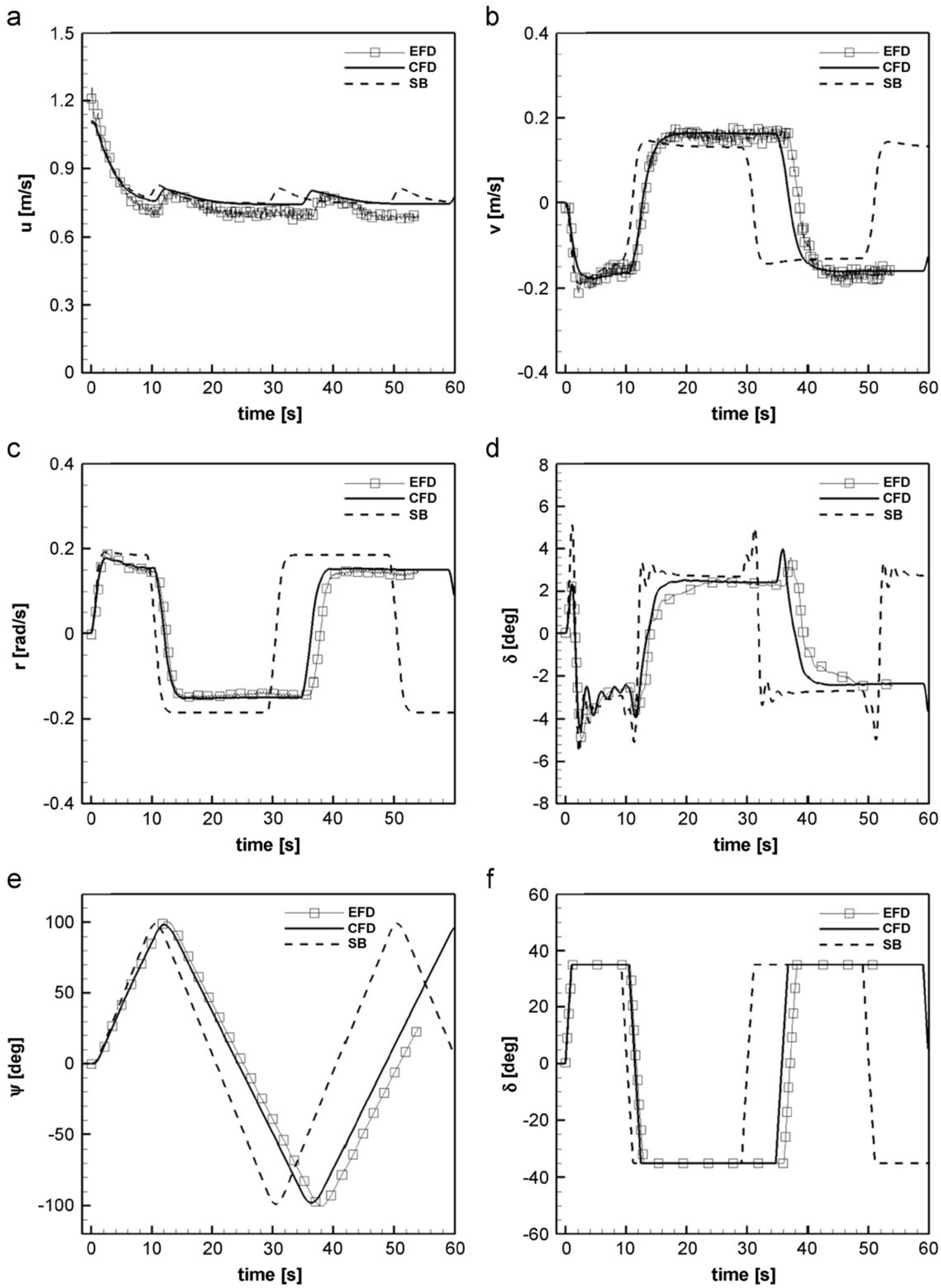

Fig. 13. Time series of large angle zigzag test (δ/ψ_c=35/90, Fr=0.20): (a) surge velocity; (b) sway velocity; (c) yaw rate; (d) roll angle; (e) yaw angle; (d) rudder angle.

Table 9
Summary of IMO criteria for all presented turning circle and zigzag cases.

Turning circle							10/10 Zigzag				20/20 Zigzag			
Parameter/Geometry	EFD	CFD	SB	EFD	CFD	SB	Parameter/Geometry	EFD	CFD	SB	Parameter/Geometry	EFD	CFD	SB
Fr	0.2			0.1			Fr	0.1			Fr	0.2		
δ (deg)	25			35			δ/ψ_c(deg)	10/10			δ/ψ_c (deg)	20/20		
Advance (A/L)	2.82	2.86	2.89	2.26	–	2.59	1st shoot angle $\alpha10_1$ (deg)	1.56	–	0.70	1st shoot angle $\alpha20_1$ (deg)	5.53	5.13	3.76
Transfer (T/L)	1.81	1.85	1.72	1.30	–	1.45	2nd shoot angle $\alpha10_2$ (deg)	1.56	–	0.71	2nd shoot angle $\alpha20_2$ (deg)	5.27	4.63	3.68
Radius (R/L)	2.17	2.20	2.12	1.62	–	1.77								
Tactical Diameter (TD/L)	4.35	4.42	4.20	3.12	–	3.55								
Criteria = A < 4.5L and TD < 5L							Criteria = $\alpha10_1 < f_{101}$(L/V) deg and $\alpha10_2 < f_{102}$(L/V) f_{101}(L/V) = 10 deg and f_{102}(L/V) = 25° for ONRT geometry at Fr = 0.1				Criteria = $\alpha20_1 < 25°$			

nonlinear steering index (*N*90) has huge error with that of EFD which indicates SB coefficients need some tuning to satisfy these large maneuvering cases. Trajectories of Z90 are shown in Fig. 10c. Again CFD shows close agreement with EFD while the SB shows large error. The time series of the EFD, CFD and SB large angle zigzag test are shown in Fig. 13. CFD and SB slightly over predict the surge velocity (Fig. 13a). CFD shows good agreement with EFD for the other state variables. SB under predicts the sway velocity (Fig. 13b) and over predicts the yaw rate (Fig. 13c) which is the main factor for the large error in the trajectory. SB over predicts the roll angle (Fig. 13d) just after the rudder deflection which is also similar as for the Z20 zigzag.

Additionally, as shown in Table 9, most of the EFD maneuvering trials and predictions by both SB and CFD adequately meet IMO criteria which show superior maneuverability of ONRT. Note that there is no criterion for large angle zigzag.

In these free-running trials, CFD shows reliability to reproduce the EFD free-running maneuvers in calm water, whereas SB shows fair ability to express the maneuvers which is inferior to the CFD. The SB errors could be reduced by tuning the hydrodynamic and rudder maneuvering coefficients using the system identification.

7. System identification results

7.1. Manoeuvring coefficients estimated by EKF

For EKF, the rudder motion and propeller RPS are input as control variables Eq. (29) and the state variables of SB, EFD and CFD free-running results are input as measurement variables Eq. (30). The added masses and added moment of inertias go to the denominator of the right hand side of Eqs. (3)–(6) as shown in Eqs. (31)–(35). Therefore the added mass and added moment of inertia values should be known to use EKF because the error of the added mass and added moment of inertia have great influence on all the other coefficients. Moreover rudder coefficients values also should be known because the rudder forces/moments cannot be separated from the state variables. Here the initial maneuvering coefficients values are set to be 120% of the original SB coefficients.

$$u(k) = \begin{bmatrix} \delta(k) & n(k) \end{bmatrix}^T \tag{29}$$

$$x(k) = \begin{bmatrix} u(k) & v(k) & p(k) & r(k) & \phi(k) \end{bmatrix}^T \tag{30}$$

$$f[(k),u(k),\Theta] = \begin{bmatrix} f_1 & f_2 & f_3 & f_4 & f_5 \end{bmatrix}^T \tag{31}$$

$$f_1 = (m+m_y)vr + T(u;n) - R(u) + X_{rr}(u)r^2 + X_{vr}(u)vr + X_{vv}(u)v^2 + X_R(\delta,u,v,r)/(m+m_x) \tag{32}$$

$$f_2 = Y_r(u)r + Y_v(u)v + Y_\phi(u)\phi + Y_{rrr}(u)r^3 + Y_{rrv}(u)r^2v + Y_{rvv}(u)rv^2 + Y_{vvv}(u)v^3 + Y_R(\delta,\phi,u,v,r) - (m+m_x)/(m+m_y) \tag{33}$$

$$f_3 = m_x z_H ur + K_r(u)r + K_p(u)p + K_v(u)v + K_\phi(u)\phi - mgGZ(\phi) + K_{rrr}(u)r^3 + K_{rrv}(u)r^2v + K_{rvv}(u)rv^2 + K_{vvv}(u)v^3 + K_R(\delta,u,v,r)/(I_x+J_x) \tag{34}$$

$$f_4 = N_r(u)r + N_v(u)v + N_\phi(u)\phi + N_{rrr}(u)r^3 + N_{rrv}(u)r^2v + N_{rvv}(u)rv^2 + N_{vvv}(u)v^3 + N_R(\delta,\phi,u,v,r)/(I_z+J_z) \tag{35}$$

$$f_5 = p(k) \tag{36}$$

Table 10 shows the ratio and difference between the original maneuvering coefficient values used in the previous SB simulations and estimated coefficients values based on EKF. Here the ratio and difference are computed by Eqs. (37) and (38). In Table 10, "Linear Av. Δ%D" is average of the Δ%D of Y_v, Y_r, K_p, N_v, and N_r. "Nonlinear Av. Δ%D" is average of the Δ%D of maneuvering coefficients excluding Y_v, Y_r, K_p, N_v, and N_r. "Total Av. Δ%D" is averaging of all the maneuvering coefficients. In Table 10, the maneuvering coefficients estimated from the SB free-running data gives small error from the original SB maneuvering coefficients which suggest that EKF has good ability to reconstruct the maneuvering coefficients from free-running trial data. However the maneuvering coefficients estimated from EFD and CFD free-running data shows large difference from the original coefficients. Especially the Y_v coefficients estimated from EFD-Z90 and CFD-Z20, Z90 free-running data have opposite sign compared to the original Y_v which has no physical meaning because the Y_v is the damping term in sway. Meanwhile the maneuvering coefficients estimated from EFD-T25 and CFD-T25 seem to have smaller difference than the maneuvering coefficients estimated form the other EFD and CFD free-running data. The global difference (Global Av. Δ%) of EFD and CFD shows similar values which is because the EFD and CFD state variables shows close agreement as shown in Figs.11–13.

$$A\,Est./Orig. = \frac{A_{estimated}}{A_{original}} \tag{37}$$

$$A\Delta\%D = \left| \frac{A_{estimated} - A_{original}}{A_{original}} \right| \times 100 \tag{38}$$

7.2. Manoeuvring coefficients estimated by CLS

CLS requires not only state variables but also total hydrodynamic forces/moments and acceleration variables during free-running trial which cannot be provided by EFD. Therefore only SB and CFD free-running data are used for CLS. In the CFD and SB data, the rudder normal force can be separated from total

Table 10
Ratio of estimated coefficients using EKF and the original coefficients: "T25" $\delta = 25^{\circ}$ turning with Fr=0.20; "Z20" δ/ψc=20/20 zigzag with Fr=0.20; "Z90" δ/ψc=90/35 large angle zigzag with Fr=0.20; "SUM" parallel processing T25, Z20 and Z90.

EKF	SB			EFD			CFD		
	T25	Z20	Z90	T25	Z20	Z90	T25	Z20	Z90
X_{vv} Est./Orig.	1.03	0.96	1.10	0.18	0.10	−0.10	1.32	2.47	2.11
X_{vr} Est./Orig.	0.97	0.97	0.95	−0.38	−0.63	−0.05	1.24	1.16	1.01
X_{rr} Est./Orig.	0.99	1.07	1.11	−0.39	−1.09	0.77	2.35	3.63	1.16
Y_{v} Est./Orig.	1.05	0.99	0.98	1.38	0.09	−0.48	1.40	−1.20	−1.73
Y_{r} Est./Orig.	1.02	0.98	0.96	1.01	−0.43	−0.41	1.58	−1.70	−1.93
Y_{vvv} Est./Orig.	0.96	1.01	0.97	1.04	0.70	1.53	1.18	1.18	1.15
Y_{vvr} Est./Orig.	1.12	1.20	1.16	1.13	1.09	1.21	1.19	1.18	1.17
Y_{vrr} Est./Orig.	0.83	0.99	0.98	0.46	0.85	0.44	1.10	0.92	0.72
Y_{rrr} Est./Orig.	1.18	0.91	1.00	0.53	1.25	1.06	1.13	0.96	0.75
Y_{ϕ} Est./Orig.	1.20	1.21	1.20	1.21	1.21	1.21	1.20	1.21	1.22
z_{H} Est./Orig.	1.00	1.00	1.00	1.51	1.22	1.17	1.31	2.08	1.51
K_{p} Est./Orig.	0.99	1.00	1.01	1.50	0.80	1.24	1.91	1.17	1.72
K_{ϕ} Est./Orig.	1.18	1.16	1.17	1.19	1.18	1.18	1.20	1.20	1.20
N_{v} Est./Orig.	1.06	0.99	1.02	0.97	2.73	2.60	1.25	2.48	1.96
N_{r} Est./Orig.	1.04	1.00	1.01	0.77	1.09	1.09	0.91	1.36	1.22
N_{vvv} Est./Orig.	1.25	1.15	1.22	0.88	1.04	1.56	1.16	1.16	1.09
N_{vvr} Est./Orig.	1.17	1.15	1.15	1.15	2.61	1.98	1.26	1.45	1.56
N_{vrr} Est./Orig.	1.17	1.25	1.16	1.95	0.22	0.39	1.22	1.04	1.07
N_{rrr} Est./Orig.	1.21	1.20	1.22	1.07	1.26	1.23	1.20	1.23	1.24
N_{ϕ} Est./Orig.	1.08	1.04	1.02	1.56	1.87	2.29	1.21	1.30	1.25
Linear Av. Δ%D	3.00	0.79	1.72	27.82	76.37	83.12	42.46	133.1	134.5
Nonlinear Av. Δ %D	11.90	10.45	10.27	48.90	65.49	54.46	26.94	43.77	25.39
Total Av. Δ %D	10.13	8.34	8.55	43.54	69.64	63.97	32.59	71.55	59.10
Global Av. Δ %D	**9.01**			**59.05**			**54.41**		

Table 11
Ratio of estimated coefficients using CLS and the original coefficients: "T25" $\delta = 25^{\circ}$ turning with Fr=0.20; "Z20" δ/ψc=20/20 zigzag with Fr=0.20; "Z90" δ/ψc=90/35 large angle zigzag with Fr=0.20; "SUM" parallel processing T25, Z20 and Z90.

CLS	SB				CFD			
	T25	Z20	Z90	SUM	T25	Z20	Z90	SUM
l_{R} Est./Orig.	1.01	1.00	0.99	0.99	0.37	0.50	1.17	0.95
γ Est./Orig.	0.96	0.97	1.00	1.01	1.43	1.14	0.72	0.79
ε Est./Orig.	1.00	1.00	1.00	0.99	0.78	0.75	0.81	0.75
m_{x} Est./Orig.	1.03	1.00	1.00	1.01	0.00	0.57	0.00	0.00
X_{vv} Est./Orig.	1.09	1.09	1.09	1.05	1.17	0.96	0.76	0.82
X_{vr} Est./Orig.	1.07	1.07	1.06	0.99	1.24	1.53	1.21	1.24
X_{rr} Est./Orig.	0.90	0.91	0.91	0.98	1.36	1.18	0.75	1.18
t_{R} Est./Orig.	1.00	1.00	1.00	1.00	0.63	0.00	0.00	0.33
m_{y} Est./Orig.	1.10	1.07	1.07	1.09	0.92	1.01	0.37	0.64
Y_{v} Est./Orig.	1.08	1.09	1.09	1.10	0.90	0.97	0.33	0.67
Y_{r} Est./Orig.	1.29	1.24	1.23	1.26	0.56	−0.24	−1.41	−0.84
Y_{vvv} Est./Orig.	1.10	1.10	1.10	1.10	0.70	0.56	0.56	1.13
Y_{vvr} Est./Orig.	1.10	1.09	1.09	1.09	−0.19	0.95	0.92	1.22
Y_{vrr} Est./Orig.	1.10	1.09	1.09	1.09	0.88	0.75	−0.40	−0.40
Y_{rrr} Est./Orig.	1.06	0.99	1.00	1.01	1.24	−0.75	0.57	0.46
Y_{ϕ} Est./Orig.	−0.94	−4.03	−3.57	1.22	8.28	9.73	−4.68	−4.68
a_{H} Est./Orig.	1.00	1.02	1.00	1.00	0.62	1.20	0.80	0.92
J_{x} Est./Orig.	1.04	1.01	0.99	0.98	0.00	0.24	0.73	2.44
z_{H} Est./Orig.	1.27	1.04	1.30	1.23	1.00	0.71	1.29	1.27
K_{p} Est./Orig.	1.10	1.11	1.04	1.04	0.82	0.49	1.24	0.82
K_{ϕ} Est./Orig.	69.1	−24.3	70.3	54.5	577.7	763.4	63.9	16.0
z_{HR} Est./Orig.	1.00	1.00	1.00	1.00	1.05	0.41	1.00	0.94
J_{z} Est./Orig.	0.97	0.96	0.97	0.99	0.72	0.63	0.82	0.76
N_{v} Est./Orig.	1.10	1.09	1.05	0.99	0.74	0.91	1.01	0.91
N_{r} Est./Orig.	1.10	1.05	1.07	1.04	0.74	0.82	0.93	0.72
N_{vvv} Est./Orig.	1.10	1.10	1.10	1.10	0.76	1.69	0.38	0.92
N_{vvr} Est./Orig.	1.10	1.10	1.10	1.10	1.30	1.11	0.95	1.28
N_{vrr} Est./Orig.	1.10	1.10	1.09	1.10	1.95	1.37	0.94	0.86
N_{rrr} Est./Orig.	1.18	1.08	1.52	1.44	2.44	1.12	1.12	1.46
N_{ϕ} Est./Orig.	1.07	0.92	1.02	1.22	1.53	5.42	3.46	1.96
x_{H} Est./Orig.	0.81	0.81	0.81	0.81	0.98	0.67	1.23	1.16
Rudder Av. Δ %D	3.40	3.36	3.03	3.06	29.98	42.98	29.70	21.21
Linear Av. Δ %D	15.85	10.26	13.08	11.38	20.66	38.93	61.41	49.82
Nonlinear Av. Δ %D	8.14	6.52	9.16	7.77	51.76	44.31	43.60	47.18
Total Av. Δ %D	8.59	6.52	8.48	7.38	38.81	39.43	40.48	39.16
Global Av. Δ %D	**7.74**				**39.47**			

hydrodynamic forces which make it possible to estimate rudder maneuvering coefficients. By using the total hydrodynamic forces/moments, the added mass and added moment of inertia can be estimated which cannot be estimated in EKF. Moreover it is expected to estimate generalized maneuvering coefficients by connecting the time series of several types of free-running trials, which is referred to as parallel processing (Abkowitz 1980). Motion continuity is not necessary for CLS which is required in EKF. Hereafter the connected data (T25, Z20, and Z90) are entitled as SUM.

The CLS attempts to minimize the square of the error between output total forces, rudder normal force Eqs. (39)–(44) and estimated forces Eqs. (45)–(49) by tuning coefficients Eqs. (50)–(54).

$$Optimize: y(k)=\min\left[\sum_{t=0}^{t_{max}}(X_{out}-X_{est}(x))^2\right] \tag{39}$$

$$X_{out1}=F_{Ntot} \tag{40}$$

$$X_{out2}=(X_{tot}-T+R) \tag{41}$$

$$X_{out3}=Y_{tot} \tag{42}$$

$$X_{out4}=(K_{tot}-mgGZ) \tag{43}$$

$$X_{out5}=N_{tot} \tag{44}$$

$$X_{est1}(x_1)=F_N=\tfrac{1}{2}\rho A_R f_\alpha U_R^2 \sin\alpha_R \tag{45}$$

$$X_{est2}(x_2)=X_{rr}(u)r^2+[X_{vr}(u)+m_y]vr+X_{vv}(u)v^2-(1-t_R)F_N\sin\delta-m_x\dot{u} \tag{46}$$

$$X_{est3}(x_3)=[Y_r(u)-m_xu]r+Y_v(u)v+Y_\phi(u)\phi+Y_{rrr}(u)r^3+Y_{rrv}(u)r^2v+Y_{rvv}(u)rv^2+Y_{vvv}(u)v^3-(1+a_H)F_N\cos\delta\cos\phi-m_y\dot{v} \tag{47}$$

$$X_{est4}(x_4)=m_xz_Hur+K_r(u)r+K_p(u)p+K_v(u)v+K_\phi(u)\phi-mgGZ(\phi)+K_{rrr}(u)r^3+K_{rrv}(u)r^2v+K_{rvv}(u)rv^2+K_{vvv}(u)v^3+z_{HR}(1+a_H)F_N\cos\delta-J_x\dot{p} \tag{48}$$

$$X_{est5}(x_5)=N_r(u)r+N_v(u)v+N_\phi(u)\phi+N_{rrr}(u)r^3+N_{rrv}(u)r^2v+N_{rvv}(u)rv^2+N_{vvv}(u)v^3-(x_R+a_Hx_H)F_N\cos\delta\cdot\cos\phi-J_z\dot{r} \tag{49}$$

$$x_1=\begin{bmatrix}\gamma_R & l_R & \varepsilon\end{bmatrix}^T \tag{50}$$

$$x_2=\begin{bmatrix}X_{rr} & (X_{vr}+m_y) & X_{vv} & t_R & m_x\end{bmatrix}^T \tag{51}$$

$$x_3=\begin{bmatrix}(Y_r-m_xu) & Y_v & Y_\phi & Y_{rrr} & Y_{rrv} & Y_{rvv} & Y_{vvv} & a_H & m_y\end{bmatrix}^T \tag{52}$$

$$x_4=\begin{bmatrix}z_H & K_p & K_\phi & z_{HR} & J_x\end{bmatrix}^T \tag{53}$$

Table 12
Steady state variables and turning geometries errors between EFD $\delta=25^\circ$ turning circle results and SB simulations using estimated coefficients: "SSV Av. E%D" average of the absolute E%D values of u, v, r, and ϕ; "ATR Av. E%D" average of the absolute E%D values of A, T, and R; "Total Av. E%D" average of "SSV Av. E%D" and "ATR Av. E%D".

	u E%D	v E%D	r E%D	ϕ E%D	SSV Av. E%D	A E%D	T E%D	R E%D	ATR Av. E%D	Total Av. E%D
EKF										
EFD-T25	9.53	5.51	10.39	20.44	11.47	−3.17	−0.73	−0.10	1.33	**6.40**
EFD-Z20	14.58	−1.36	10.70	7.80	8.61	−3.26	−6.82	−4.05	4.71	**6.66**
EFD-Z90	1.01	21.68	14.45	−6.46	10.90	3.58	13.96	10.20	9.25	**10.07**
CFD-T25	−6.03	−10.37	−2.55	−16.17	8.78	−2.02	1.93	2.69	2.22	**5.50**
CFD-Z20	−11.41	−17.90	−1.25	17.57	12.03	3.61	9.26	9.86	7.58	**9.80**
CFD-Z90	1.01	21.68	14.45	−6.46	10.90	3.58	13.96	10.20	9.25	**10.07**
CLS										
CFD-T25	3.63	−3.53	0.55	−8.17	3.97	−8.28	−5.88	−4.01	6.05	**5.01**
CFD-Z20	0.44	0.26	12.99	−9.84	5.88	0.62	8.22	9.79	6.21	**6.05**
CFD-Z90	5.31	0.86	−2.48	−3.30	2.99	−11.48	−11.55	−9.19	10.74	**6.86**
CFD-SUM	5.04	−3.64	−0.47	−4.35	3.38	−8.63	−9.39	−6.52	8.18	**5.78**

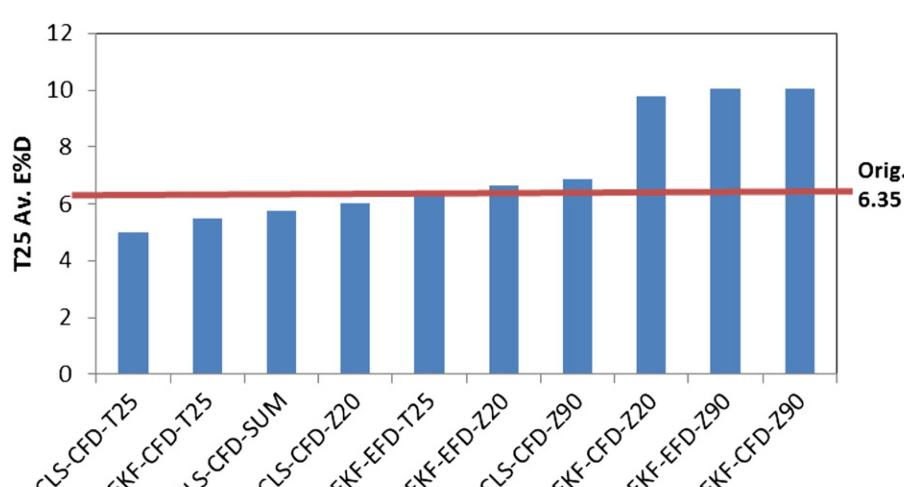

Fig. 14. Total average errors of T25 SB simulations using estimated and original maneuvering coefficients. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 13
Steering indices and overshoot angle errors between EFD 20/20 zigzag results and SB simulations using estimated coefficients: "KTN Av. E%D" average of the absolute E%D values of K20, T20, and N20; "OS Av. E%D" average of the absolute E%D values of 1st OS and 2nd OS; "Total Av. E%D" average of "KTN Av. E%D" and "OS Av. E%D".

	K20 E%D	T20 E%D	N20 E%D	KTN Av. E%D	1st OS[a] E%D	2nd OS E%D	OS Av. E%D	Total Av. E%D
EKF								
EFD-T25	−0.47	−30.31	11.19	13.99	−5.18	−4.33	4.76	**9.37**
EFD-Z20	7.81	−7.98	19.68	11.82	−2.13	−0.91	1.52	**6.67**
EFD-Z90	22.45	22.44	−100.00	48.30	2.60	3.32	2.96	**25.63**
CFD-T25	−5.94	−27.69	−4.21	12.62	−5.49	−5.23	5.36	**8.99**
CFD-Z20	−2.23	33.62	17.04	17.63	−1.33	−1.51	1.42	**9.53**
CFD-Z90	24.26	21.22	−100.00	48.49	2.60	3.32	2.96	**25.73**
CLS								
CFD-T25	−10.28	−22.56	−16.53	16.46	−6.11	−5.67	5.89	**11.17**
CFD-Z20	−0.39	3.01	0.17	1.19	−2.26	−1.46	1.86	**1.52**
CFD-Z90	−10.99	−16.54	−16.78	14.77	−5.76	−4.89	5.32	**10.05**
CFD-SUM	−6.49	−9.59	−10.94	9.01	−4.34	−3.56	3.95	**6.48**

[a] OS; overshoot angle.

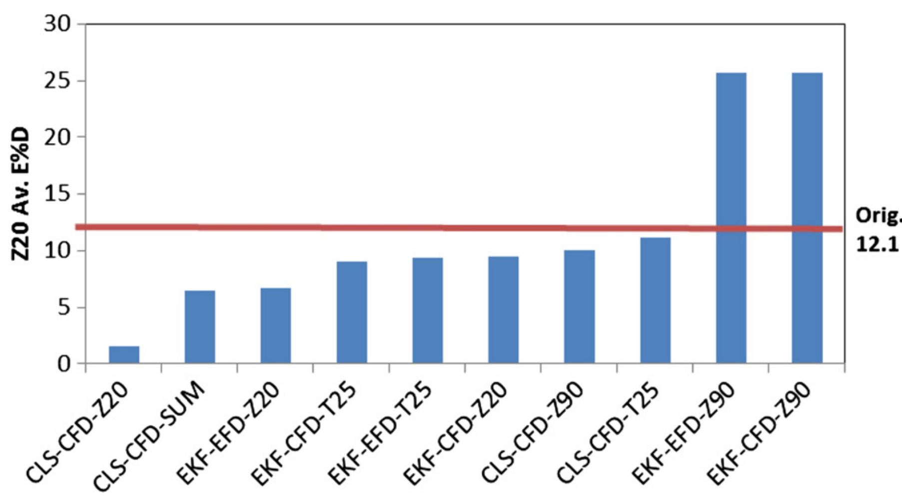

Fig. 15. Total average errors of Z20 SB simulations using estimated and original maneuvering coefficients. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 14
Steering indices and overshoot angle errors between EFD 90/35 large angle zigzag results and SB simulations using estimated coefficients: "KTN Av. E%D" average of the absolute E%D values of K90, T90, and N90; "OS Av. E%D" average of the absolute E%D values of 1st OS and 2nd OS; "Total Av. E%D" average of "KTN Av. E%D" and "OS Av. E%D".

	K90 E%D	T90 E%D	N90 E%D	KTN Av. E%D	1st OS E%D	2nd OS E%D	OS Av. E%D	Total Av. E%D
EKF								
EFD-T25	20.05	−36.64	−100.00	52.23	−0.37	−0.66	0.52	**26.37**
EFD-Z20	5.82	−28.46	−13.62	15.97	−0.84	−1.14	0.99	**8.48**
EFD-Z90	7.42	−7.80	−4.26	6.49	−0.26	−0.68	0.47	**3.48**
CFD-T25	0.37	−38.92	−16.52	18.60	−1.90	−2.14	2.02	**10.31**
CFD-Z20	−5.15	−35.40	−20.71	20.42	−2.36	−2.57	2.47	**11.44**
CFD-Z90	7.42	−7.80	−4.26	6.49	−0.26	−0.68	0.47	**3.48**
CLS								
CFD-T25	31.97	10.44	−100.00	47.47	2.16	1.93	2.04	**24.76**
CFD-Z20	40.22	41.62	−100.00	60.61	2.76	2.72	2.74	**31.68**
CFD-Z90	2.26	−11.46	−9.33	7.68	−1.23	−1.49	1.36	**4.52**
CFD-SUM	2.98	−9.87	−10.37	7.74	−0.98	−1.25	1.12	**4.43**

$$x_5 = \begin{bmatrix} N_r & N_v & N_\phi & N_{rrr} & N_{rrv} & N_{rvv} & N_{vvv} & x_H & J_z \end{bmatrix}^T \quad (54)$$

Table 11 shows the ratio and difference between the original maneuvering coefficients including rudder maneuvering coefficients values used in the SB simulations and the maneuvering coefficients values estimated by CLS. Here the EFD free-running data cannot be applied to the CLS method because the EFD acceleration terms are usually too noisy to use and this method requires rudder normal force during the free-running test. However there is no interest to use EFD free-running data since the CFD can reproduce the EFD free-running results as shown in Figs. 9–13.

In Table 11, "Rudder Av. Δ%D" is average of the Δ%D of rudder maneuvering coefficients l_R, γ_R, ε, t_R, a_H, z_{HR}, and x_R. Here Δ%D of the maneuvering coefficients related to ϕ (Y_ϕ, K_ϕ, and N_ϕ) are excluded from the averaging Δ%D. This is because the errors of ϕ maneuvering coefficients are extremely large. In fact, the effect of the ϕ maneuvering coefficients are too small to compute correctly in calm water maneuvers and the original values of ϕ maneuvering coefficients values are also very small which causes the errors to be very large. Also it is impossible to distinguish the restoring term $mgGZ(\phi)$ and $K_\phi\phi$ term in this model which means the error of $mgGZ(\phi)$ directly goes to the K_ϕ.

In Table 11, the estimated maneuvering coefficients using SB simulations shows less than 8% error from the original SB maneuvering coefficients in total which suggest that the CLS has good ability to reconstruct the maneuvering coefficients from the free-running trial data. The constraints ensure that the sign of the estimated linear maneuvering coefficients are correct in Table 11.

8. Disscussion on the system identification results

By applying system EKF and CLS identification techniques, 10 sets of maneuvering coefficients are estimated from the several CFD and EFD free-running data (T25, Z20, Z90, and SUM). However, by just examining Tables 10 and 11, it is difficult to conclude which set of maneuvering coefficients are the best ones which can cover the widest range of maneuvers. The SB simulation using the original coefficients have some errors with EFD and CFD free-running results as shown in Figs. 9–13. Therefore the set of original maneuvering coefficients are not optimum. Hence SB simulations using the estimated maneuvering coefficients are executed clarify which set of estimated maneuvering coefficients are the most generalized ones for the SB mathematical model. Here the 10 sets of coefficients are labeled to simplify the discussions. For example, a set of maneuvering coefficients named "CLS-CFD-Z90" indicates that these maneuvering coefficients are estimated by CLS using CFD-Z90 free-running data.

First 10 sets of $T25$ are computed by SB simulation using the 10 sets of estimated maneuvering coefficients. Table 12 shows the errors between EFD-T25 results and the SB simulations using estimated maneuvering coefficients. The error criteria follow Table 6. Fig. 14 shows the "Total Av. E%D" errors shown in Table 12 sorted by the value of "Total Av. E%D". The Red line shows the "Total Av. E%D" of the original SB simulation shown in Table 6. As expected, the maneuvering coefficients estimated by T25 (CLS-CFD-T25, EKF-CFD-T25, CLS-CFD-SUM, and EKF-EFD-T25) gives smaller error than that of using other data (Z20 and Z90). Most of the maneuvering coefficients using the Z20 and Z90 data give larger error than the original maneuvering coefficients.

Table 13 shows the errors between EFD-Z20 results and the SB-Z20 simulations using estimated maneuvering coefficients. The error criteria follow Table 7. Fig. 15 shows the "Total Av. E%D" errors shown in Table 13 sorted by the value of "Total Av. E%D". The Red line shows the "Total Av. E%D" of the original SB simulation shown in Table 7. In Fig. 15, most of the estimated maneuvering coefficients show better results than the original maneuvering coefficients. Especially CLS-CFD-Z20 shows small error. Table 14 shows the errors between EFD-Z90 results and the SB-Z90 simulations using estimated maneuvering coefficients. Fig. 16 shows the "Total Av. E%D" errors shown in Table 14 sorted by the value of "Total Av. E%D". The Red line shows the "Total Av. E%D" of the original SB simulation shown in Table 8. In Fig. 16, the maneuvering coefficients estimated by Z90 (EKF-EFD-Z90, EKF-CFD-Z90, CLS-CFD-SUM, and CLS-CFD-Z90) show much smaller error than original maneuvering coefficients.

Table 15 shows the "Total Av. E%D" of T25, Z20, and Z90 and the "Global Av. E%D" which are averages of T25, Z20, and Z90's "Total Av. E%D". Fig. 17 shows "Global Av. E%D" of Table 15 sorted

Table 15
Average errors between EFD free-running (T25, Z20, Z90) results and SB simulations using estimated coefficients: "T25 E%D" total average E%D of Table 12; "Z20 E%D" total average E%D of Table 13; "Z90 E%D" total average E%D of Table 14; "Global Av. E%D" average of "T25 Av. E%D", "Z20 Av. E%D", and "Z90 Av. E%D".

	T25 E%D	Z20 E%D	Z90 E%D	Global Av. E%D
EKF				
EFD-T25	6.40	9.37	26.37	**14.05**
EFD-Z20	6.66	6.67	8.48	**7.27**
EFD-Z90	10.07	25.63	3.48	**13.06**
CFD-T25	5.50	8.99	10.31	**8.27**
CFD-Z20	9.80	9.53	11.44	**10.26**
CFD-Z90	10.07	25.73	3.48	**13.09**
CLS				
CFD-T25	5.01	11.17	24.76	**13.65**
CFD-Z20	6.05	1.52	31.68	**13.08**
CFD-Z90	6.86	10.05	4.52	**7.14**
CFD-SUM	5.78	6.48	4.43	**5.56**

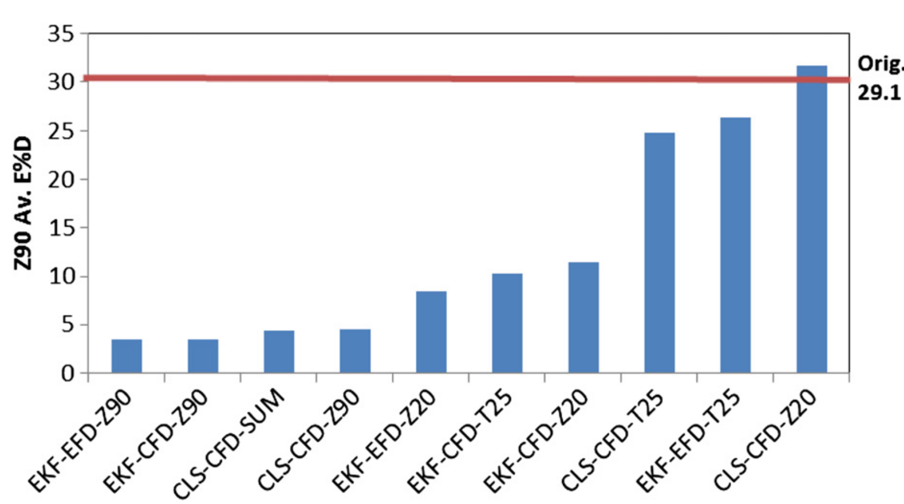

Fig. 16. Total average errors of Z90 SB simulations using estimated and original maneuvering coefficients. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

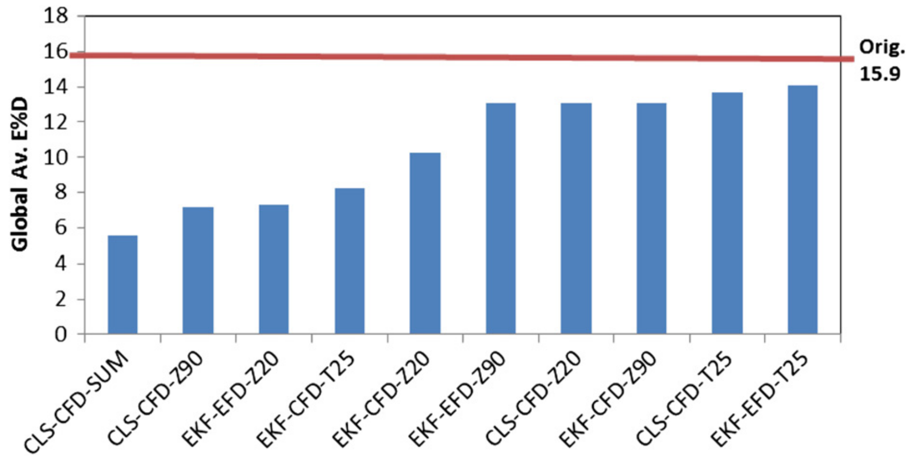

Fig. 17. Global average errors of T25, Z20, and Z90 SB simulations using estimated and original maneuvering coefficients. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

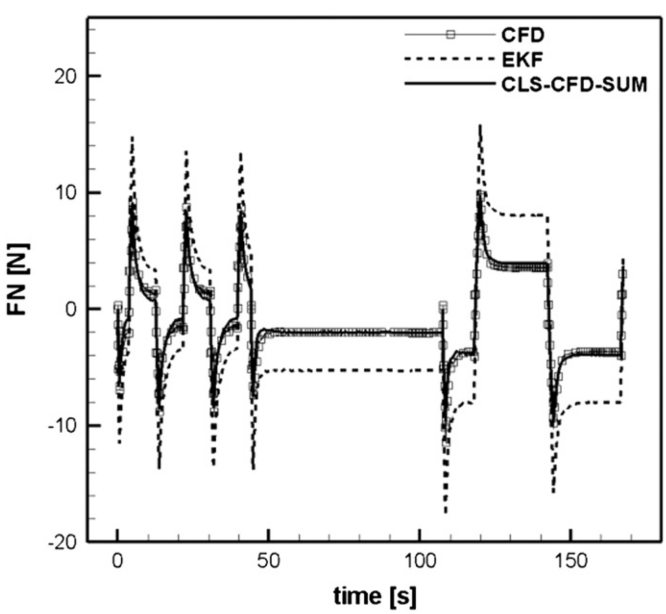

Fig. 18. Comparison of the rudder normal forces between CFD and SB simulations (EKF and CLS-CFD-SUM).

Table 16
Errors between blind EFD free-running results ($\delta=35^{\circ}$ turning circle and 10/10 zigzag tests with Fr=0.10) and SB simulations using original and CLS-CFD-SUM coefficients: "T35Fr010 Av. E%D" average of "SSV Av. E%D" and "ATR Av. E%D"; "Z10Fr010 Av. E%D" average of "KTN Av. E%D" and "OS Av. E%D"; "Global Av. E%D" average of "T35Fr010 Av. E%D" and "Z10Fr010 Av. E%D".

	SB-Orig.	CLS-CFD-SUM
$\delta=35^{\circ}$ Turning with Fr=0.10		
u E%D	−11.81	7.72
v E%D	−8.36	32.84
r E%D	−15.41	9.67
ϕ E%D	−36.91	28.84
SSV Av. E%D	18.12	19.77
A E%D	−14.83	−6.22
T E%D	−12.29	−5.11
R E%D	−8.94	−4.01
ATR Av. E%D	12.02	5.11
T35Fr010 Av. E%D	15.07	12.44
10/10 Zigzag with Fr=0.10		
K10 E%D	−20.08	0.63
T10 E%D	−28.85	4.41
N10 E%D	−30.13	−8.05
KTN Av. E%D	26.35	4.37
1st OS E%D	−7.47	−3.45
2nd OS E%D	−7.27	−3.13
OS Av. E%D	7.37	3.29
Z10Fr010 Av. E%D	16.86	3.83
Global Av. E%D	**15.97**	**8.13**

by the value of "Global Av. E%D". The Red line shows the "Global Av. E%D" of the SB simulations using the original maneuvering coefficients. In Fig. 17, all the maneuvering coefficients show smaller error than original maneuvering coefficients. Especially CLS-CFD-SUM shows the best results overall. As shown in Table 15, although CLS-CFD-SUM does not show the smallest error in each case, it has small error in all cases which brings CLS-CFD-SUM as the best maneuvering coefficients in all. Some of the other maneuvering coefficients (CLS-CFD-Z90 and EKF-EFD-Z20) also show small errors.

In the EKF method, empirical values are used for the rudder model. Fig. 18 shows the comparison of the rudder normal force between CFD and the SB simulations with the same imposed model and rudder motions. The SB simulation using "CLS-CFD-SUM" maneuvering coefficients shows close agreement with CFD while the SB simulation using EKF maneuvering coefficients strongly overestimates the rudder force. Therefore, in EKF method, the errors of the rudder forces are dominated by the hull forces and mix-up the maneuvering coefficients to balance with the ship motions. This could be a reason that some of the Y_v maneuvering coefficients show opposite sign in the EKF estimates in Table 10. Hence, even the ship motion seems to be fair, the appropriateness of the EKF maneuvering coefficients is doubtful as a physical model. The CLS seems to be fair to estimate reasonable maneuvering coefficients by treating the forces/moments directly. To predict reasonable maneuvering coefficients for general ship maneuvers, several trials should be used with parallel processing (CFD-SUM) or several kinds of motion should be taken into one trial such as the Z90 test.

The "CLS-CFD-SUM" maneuvering coefficients, which shows the best generalization in all 10 maneuvering coefficients sets, are applied for other types of free-running trials: $\delta=35^{\circ}$ turning and $\delta/\psi_c=10/10$ zigzag with different model speed Fr=0.10. These trials were not included in the previous system identification. Fig. 19a shows the trajectories of $\delta=35^{\circ}$ turning with Fr=0.10.

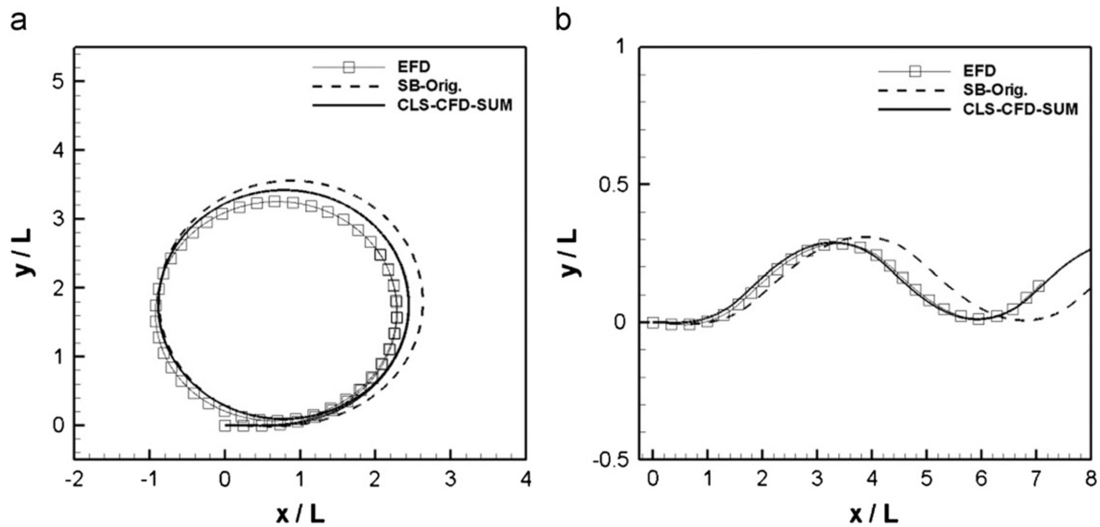

Fig. 19. Trajectories of free-running simulations: (a) turning circle ($\delta=35^\circ$, Fr=0.10); (b) zigzag ($\delta/\psi_c=10/10$, Fr=0.10).

The trajectory of the SB simulation using "CLS-CFD-SUM" maneuvering coefficients shows better agreement with EFD than that of SB simulation using the original maneuvering coefficients (SB-Orig.). The "CLS-CFD-Comb" simulation also shows better agreement in steady state variables and turning geometries during the turning test except sway velocity as shown in Table 16. The error of sway velocity makes the gap in the trajectory in Fig. 15a. Nevertheless the "CLS-CFD-SUM" shows better agreement than the "SB-Orig." in total.

Fig. 19b shows the trajectories of $\delta/\psi_c=10/10$ zigzag with Fr=0.10. The trajectory of SB simulation using "CLS-CFD-SUM" maneuvering coefficients shows close agreement with EFD while trajectory of "SB-Orig." shows some discrepancy with EFD. In the time series Fig. 20, the surge velocity Fig. 20a of "CLS-CFD-SUM" shows some error from EFD but it is negligibly small and does not influence the total model motion. "CLS-CFD-SUM" shows better agreement with EFD than that of "SB-Orig." in the other state variables Fig. 20b–d. Especially "CLS-CFD-SUM" yaw rate shows nearly perfect agreement with EFD while "SB-Orig." under estimates it. The errors of steering indices and overshoot angles are shown in Table 16. "CLS-CFD-SUM" shows better agreement in both steering indices and overshoot angles than "SB-Orig." results.

The set of "CLS-CFD-SUM" maneuvering coefficients shows better ability to predict a wide range of maneuvers than "SB-Orig." maneuvering coefficients. For these reasons, it could be possible to replace the numerous captive model tests with several CFD free-running trials using the system identification technique which brings more generalized maneuvering coefficients. Table 11 shows that the differences between the "CLS-CFD-SUM" and original SB maneuvering coefficients are significant: for the linear maneuvering coefficients the average differences are 49.8%D and the maximum is -368%D for Y_ϕ; for the nonlinear maneuvering coefficients the average differences are 47.2%D and the maximum is -46%D for N_{rrr}; some of the maneuvering coefficients (Y_r, Y_ϕ, Y_{vrr}) are estimated with different signs; and the estimated rudder coefficients are 21% different compared to the original values.

9. Conclusions

System identification methods using CFD free-running trial data is shown to be the most accurate and efficient approach for estimating the maneuvering coefficients in the SB mathematical model. Hydrodynamic and rudder maneuvering coefficients included in a 4DOF SB model are estimated from only a few CFD free-running simulations trial data employing EKF and CLS system identification methods.

Free-running tests in calm water are executed in the IIHR wave basin for the CFD and SB validations. CFD simulation results shows close agreement with the EFD which suggest the CFD simulation can reproduce the EFD free-running results. SB simulations using 4DOF nonlinear maneuvering mathematical model, which include maneuvering coefficients from numerous captive model experiments and empirical rudder maneuvering coefficients, shows fair ability to predict calm water maneuvers within some errors.

With these several free-running data, the maneuvering coefficients included in the SB model are estimated by EKF and CLS system identification techniques. Both EKF and CLS show ability to estimate the EKF and CLS coefficients from free-running data. EKF is useful to estimate maneuvering coefficients from the EFD free-running data because the EKF only needs state variables for the computation. However EKF has a basic problem to estimate rudder forces which disturbs estimating the other maneuvering coefficients. On the other hand, CLS can estimate not only the maneuvering coefficients but also the rudder maneuvering coefficients which predicts the rudder force correctly and the constraints ensure that the maneuvering coefficients have reasonable values. For CLS, the CFD free-running simulations are necessary because the CFD simulations can provide both total hydrodynamic forces/moments and local rudder normal forces. By parallel processing several free-running data, the estimated maneuvering coefficients become more generalized than using just one free-running data which indicates that the maneuvering coefficients are appropriate for a wide range of maneuvers. Especially the SB simulation using the set of "CLS-CFD-SUM" coefficients shows better agreement with EFD than using the original maneuvering coefficients even for other types of free-running trials which were not included in the system identification.

Hence, by using the present approach, several CFD free-running simulations can replace the many number of captive model experiments needed to estimate the maneuvering coefficients for the SB model. Furthermore reliable maneuvering simulations could be possible without any model experiments within short computation time. Note that similar approach is

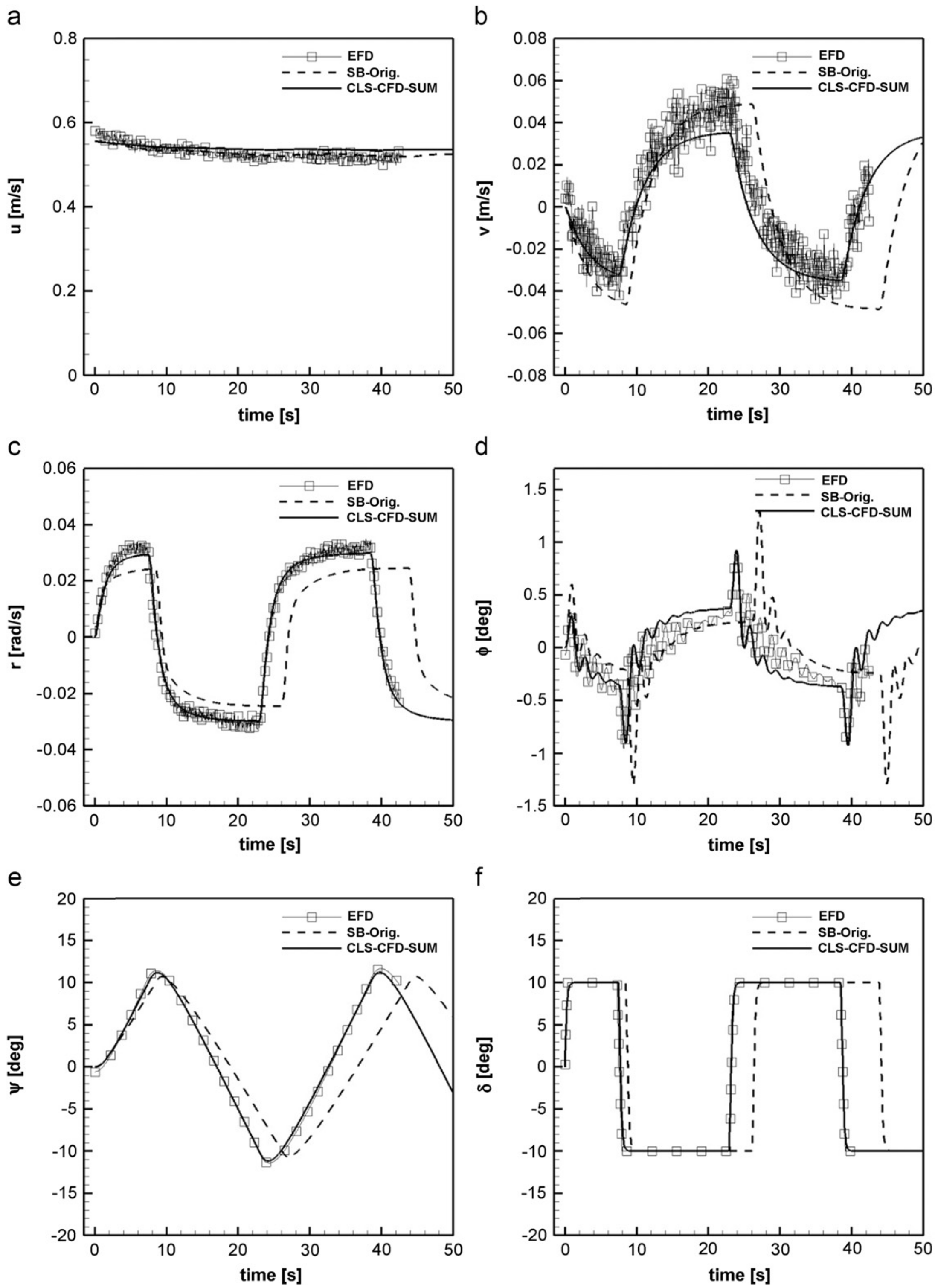

Fig. 20. Time series of zigzag test ($\delta / \psi_c = 10/10$, Fr=0.10): (a) surge velocity; (b) sway velocity; (c) yaw rate; (d) roll angle; (e) yaw angle; (d) rudder angle.

currently under development in aerospace, but using CFD to replace captive model EFD tests (Morton, 2011).

The current SB model for wave conditions (Umeda et al., 2008 and Yasukawa, 2006) have some problems for quantitative agreement with EFD while CFD free-running simulation in waves shows quantitative agreement. Therefore the SB wave model should be validated and modified by using CFD free-running simulations as future work. Moreover full-scale maneuvering coefficients can be provided by CFD full-scale simulation since the CFD can be extended to full-scale model.

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Appendix A. Extended Kalman filtering

The nonlinear system can be expressed with the following set of equations

$$\dot{x}(k) = f[x(k), u(k), \Theta] + w(k) \tag{A.1}$$

were $x(k)$ is true state variables, $u(k)$ is control variables and $w(k)$ is process noise. Θ is the vector of unknown parameters given by

$$\Theta = [x_0, \beta] \tag{A.2}$$

were, x_0 represent value of the state variables at time $k=0$ and β represent parameters in the mathematical model that defined the system characteristics.

Measurement equation can be expressed with Eqs. (A.3) and (A.4).

$$y(k) = h[x(k), u(k), \Theta] \tag{A.3}$$

$$z(k) = y(k) + v(k) \tag{A.4}$$

where $z(k)$ is measurement variables affected by h, the measurement model which maps the true state space into the measurement space, and measurement noise $v(k)$.

The following assumptions are made on the process noise $w(k)$ and measurement noise $v(k)$:

$$E[w(k)] = 0 : Cov[w(k)] = Q \tag{A.5}$$

$$E[v(k)] = 0 : Cov[v(k)] = R \tag{A.6}$$

In the above, it is assumed that the noises are zero-mean and white Gaussian with Q and R as the covariance matrix of these noises.

The new augmented state vector is

$$x_a = \begin{bmatrix} x & \Theta \end{bmatrix}^T \tag{A.7}$$

$$\dot{x}(k) = \begin{bmatrix} f(x_a, u, t) \\ 0 \end{bmatrix} + \begin{bmatrix} G \\ 0 \end{bmatrix} w(k) \tag{A.8}$$

$$\dot{x}(k) = f_a(x_a, u, t) + G_a w(k) \tag{A.9}$$

$$y(k) = h_a[x_a, u, t] \tag{A.10}$$

$$z_m(k) = y(k) + v(k), \quad k = 1, \ldots, N \tag{A.11}$$

here

$$f_a(t) = \begin{bmatrix} f & 0 \end{bmatrix}^T \tag{A.12}$$

$$h_a(t) = \begin{bmatrix} h & 0 \end{bmatrix}^T \tag{A.13}$$

$$G_a = \begin{bmatrix} G & 0 \end{bmatrix}^T \tag{A.14}$$

The estimation algorithm is obtained by linearizing Eqs. (A.1) and (A.2) around the prior/current best estimate of the state at each time and then applying the Kalman filtering algorithm to the linearized model. The linearized system matrices are defined as

$$A(k) = \left.\frac{\partial f_a}{\partial x_a}\right|_{x_a = \hat{x}_a(k), u = u(k)} \tag{A.15}$$

$$H(k) = \left.\frac{\partial h_a}{\partial x_a}\right|_{x_a = \tilde{x}_a(k), u = u(k)} \tag{A.16}$$

And the state transition matrix is given by

$$\phi(k) = \exp[-A(k)\Delta t] \tag{A.17}$$

where

$$\Delta t = t(k+1) - t(k) \tag{A.18}$$

The filtering algorithm is given in two parts: (i) time propagation and (ii) measurement update. In the above equations, we notice the time-varying nature of A, H and ϕ, since they are evaluated at the current state estimate, which varies with time k.

(i) Time propagation
The states are propagated from the present state to the next time instant.
The predicted state is given by

$$\tilde{x}_a(k+1) = \hat{x}_a(k) + \int_{t_k}^{t_{k+1}} f_a(\hat{x}_a, u, t)\,dt \tag{A.19}$$

In the absence of knowledge of process noise, Eq. (A.19) gives the predicted estimate of the state based on the initial/current estimate. The Runge–Kutta integrations are used in Eq. (A.19). The covariance matrix for state error (here state is x_a) propagate from instant k to $k+1$ as

$$\tilde{P}(k+1) = \phi(k)\hat{P}(k)\phi^T(k) + G_a(k)Q(k)G_a^T(k) \tag{A.20}$$

Here $\tilde{P}(k+1)$ is the predicted covariance matrix for the instant $k+1$, G_a is the process noise related coefficient matrix, and Q is the process noise covariance matrix.

(ii) Measurement update
The extended Kalman filter updates the predicted estimates by incorporating the measurement as and when they become available as follow:

$$\hat{x}_a(k+1) = \tilde{x}_a(k+1) + K(k+1)z_m(k+1) - h_a[\tilde{x}_a(k+1), u(k+1), t(k)] \tag{A.21}$$

here, K is the Kalman gain matrix.

The covariance matrix is updated using the Kalman gain and the linearized measurement matrix from the predicted covariance matrix $\tilde{P}(k+1)$.

The Kalman gain expression is given as

$$K(k+1) = \tilde{P}(k+1)H^T(k+1)[H(k+1)\tilde{P}(k+1)H^T(k+1) + R]^{-1} \tag{A.22}$$

A posteriori covariance matrix expression is given as

$$\hat{P}(k+1) = [I - K(k+1)H(k+1)]\tilde{P}(k+1) \tag{A.23}$$

Here the UD factorization (Thornton and Bierman, 1980) can also be conveniently used in the EKF, since Eqs. (A.22) and (A.23) can be put in the factorization form and processed, to avoid the drifting problem.

Appendix B. Generalized reduced gradient argorithm

The subject equations are modified to equality equations by adding slack variables.

$$\text{Optimize} : y(x)$$
$$\text{Subject to} : f_i(x) = 0 \; for \; i = 1, 2, \ldots, m \tag{B.1}$$

To develop this method the independent variables are separated into basic and nonbasic ones. There are m basic variables x_b, and (n–m) nonbasic variables x_{nb}, i.e.:

$$f_i(x) = f_i(x_b, x_{nb}) = 0 \text{ for } i = 1,2,\ldots,m \tag{B.2}$$

Indicating the solution of x_b in terms of x_{nb} from Eq. (B.2) gives:

$$x_{b,i} = \tilde{f}_i(x_{nb}) \text{ for } i = 1,2,\ldots,m \tag{B.3}$$

In nonlinear programming, the nonbasic variables are used to compute the values of the basic variable and are manipulate to obtain the optimum of the model. The model can be thought of as a function of the nonbasic variables only, that is if the constraint Eq. (B.3) are used to eliminate the basic variables, i.e.:

$$y(x) = y(x_b, x_{nb}) = y[\tilde{f}_i(x_{nb}), x_{nb}] = Y(x_{nb}) \tag{B.4}$$

Expanding the above equation in a Taylor series about x_k and including only the first order terms gives:

$$\sum_{j=1}^{m} \frac{\partial y(x_k)}{\partial x_{b,j}} dx_{b,j} + \sum_{j=1}^{n} \frac{\partial y(x_k)}{\partial x_{nb,j}} dx_{nb,j} = \sum_{j=1}^{n} \frac{\partial Y(x_k)}{\partial x_{nb,j}} dx_{nb,j} \tag{B.5}$$

In matrix notation Eq. (B.5) can be written as

$$\nabla^T Y(x_k) dx_{nb} = \nabla^T y_b(x_k) dx_b + \nabla^T y_{nb}(x_k) dx_{nb} \tag{B.6}$$

A Taylor series expansion of the constraint Eq. (B.2) gives an equation that can be substitute into Eq. (B.6) to eliminate the basic variables.

$$\sum_{j=1}^{m} \frac{\partial f_i(x_k)}{\partial x_j} dx_{b,j} + \sum_{j=1}^{m} \frac{\partial f_i(x_k)}{\partial x_j} dx_{nb,j} = 0 \text{ for } i = 1,2,\ldots,m \tag{B.7}$$

Or in matrix for Eq. (B.7) is:

$$\begin{pmatrix} \frac{\partial f_1(x_k)}{\partial x_1} & \cdots & \frac{\partial f_1(x_k)}{\partial x_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(x_k)}{\partial x_1} & \cdots & \frac{\partial f_m(x_k)}{\partial x_m} \end{pmatrix} \begin{pmatrix} dx_{b,1} \\ \vdots \\ dx_{b,m} \end{pmatrix} + \begin{pmatrix} \frac{\partial f_1(x_k)}{\partial x_{m+1}} & \cdots & \frac{\partial f_1(x_k)}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m(x_k)}{\partial x_{m+1}} & \cdots & \frac{\partial f_m(x_k)}{\partial x_n} \end{pmatrix} \begin{pmatrix} dx_{nb,m+1} \\ \vdots \\ dx_{nb,n} \end{pmatrix} = 0 \tag{B.8}$$

The following equation defines B_b as the matrix of f_i associated with the basic variables and B_{nb} as the matrix associated with the nonbasic variables, i.e.:

$$B_b dx_b + B_{nb} dx_{nb} = 0 \tag{B.9}$$

This is a convenient form of Eq. (B.8) which can be used to eliminate dx_b from Eq. (B.6). Solving Eq. (B.9) for dx_b gives:

$$dx_b = -B_b^{-1} B_{nb} dx_{nb} \tag{B.10}$$

Substituting Eq. (B.10) into (B.6) gives:

$$\nabla^T Y(x_k) dx_{nb} = -\nabla^T y_b(x_k) B_b^{-1} B_{nb} dx_{nb} + \nabla^T y_{nb}(x_k) dx_{nb} \tag{B.11}$$

Eliminating dx_{nb} from Eq. (B.11), the equation for the reduced gradient $Y(x_k)$ is obtained.

$$\nabla^T Y(x_k) = -\nabla^T y_b(x_k) B_b^{-1} B_{nb} + \nabla^T y_{nb}(x_k) \tag{B.12}$$

Knowing the value of the first partial derivatives of the economic model and constraint equations at a feasible point, the reduced gradient can be computed by Eq. (B.12). This will satisfy the model and the constraint equations. The generalized reduced gradient method use the reduced gradient to locate better value of the model in the same way as unstrained gradient search using Lagrange multipliers, i.e.:

$$x_{nb,k+1} = x_{nb,k} + a \nabla^T Y(x_k) \tag{B.13}$$

where is the parameter of the line along the reduced gradient. A line search on is used to locate the optimum of $Y(x_{nb})$ along the reduced gradient line from x_k.

In taking trial step as a is varied along the generalized reduced gradient line, the matrices B_b and B_{nb} must be evaluated along with the gradients $\nabla^T y_b(x_b)$ and $\nabla^T y_{nb}(x_k)$. This requires knowing both x_{nb} and x_b at each step. The values of x_{nb} are obtained from (B.13). However, Eq. (B.2) must be solved for x_b; and frequency, this must be done numerically using the Newton–Raphson method. The Newton–Raphson algorithm in terms of the nomenclature for this procedure is given by the following equation.

$$x_{b,i+1} = x_{b,i} - B_b^{-1} f_i(x_{b,i}, x_{nb,i}) \tag{B.14}$$

Where the value of the root of the constraint Eq. (B.2) are being sought for x_b, having computed x_{nb} from Eq. (B.13). Thus, the derivatives computed for the generalized reduced gradient's B_b matrix can be used in the Newton–Raphson root seeking procedure also.

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